Lecture 02

Market Failures

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AEM 4510

Roadmap

- What are market failures?
- When do they happen?
- What are the consequences?

Market failures and the environment

The ideal world: graphical

In the best case scenario, a market equilibrium leads to the efficient allocation

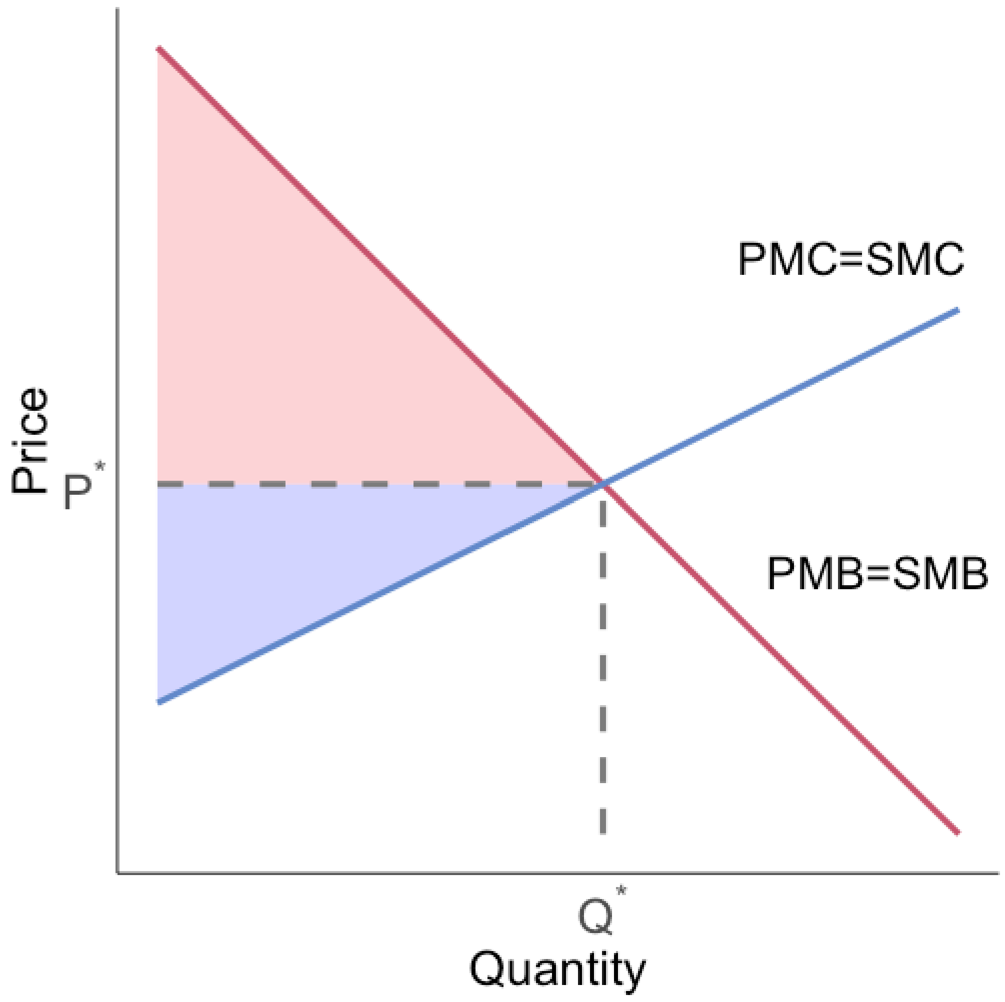
We have a private bread supply curve (private MC)

We have a private bread demand curve (private MB)

In equilibrium: supply = demand so PMC = PMB = price

For bread, the private costs and benefits are very likely the social costs and benefits

Market equilibrium

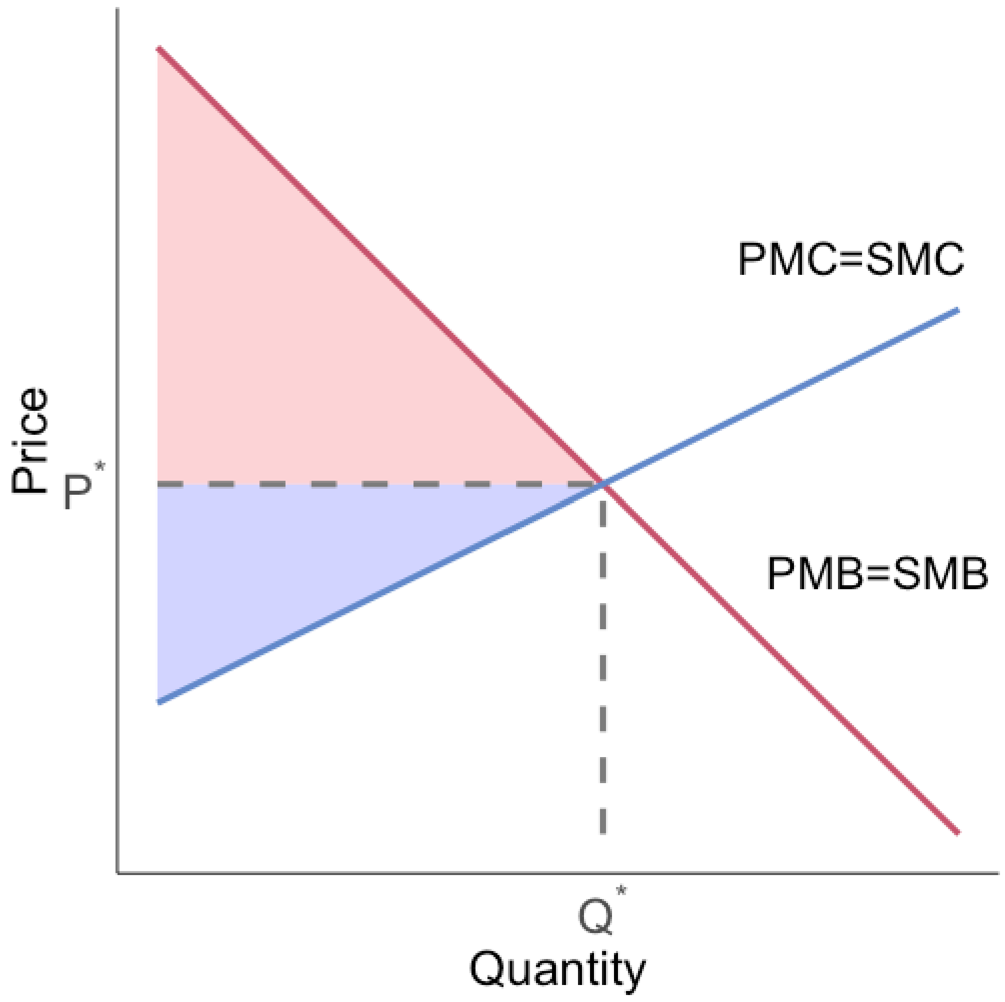


Consumer surplus is the difference between willingness to pay (demand) and price

Producer surplus is the difference between price and marginal cost (supply)

Total surplus is the sum of CS and PS

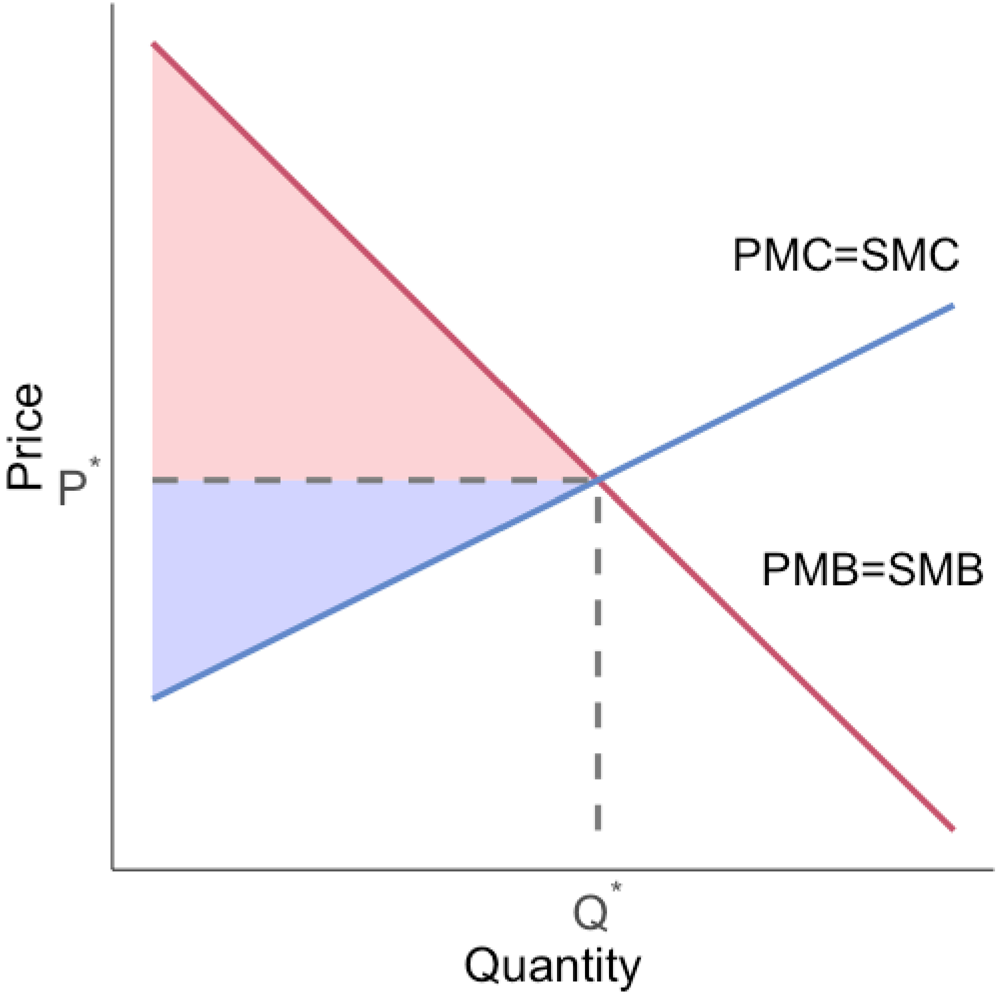
Market equilibrium



For bread, the private costs and benefits are very likely the social costs and benefits

What does this mean about the market allocation?

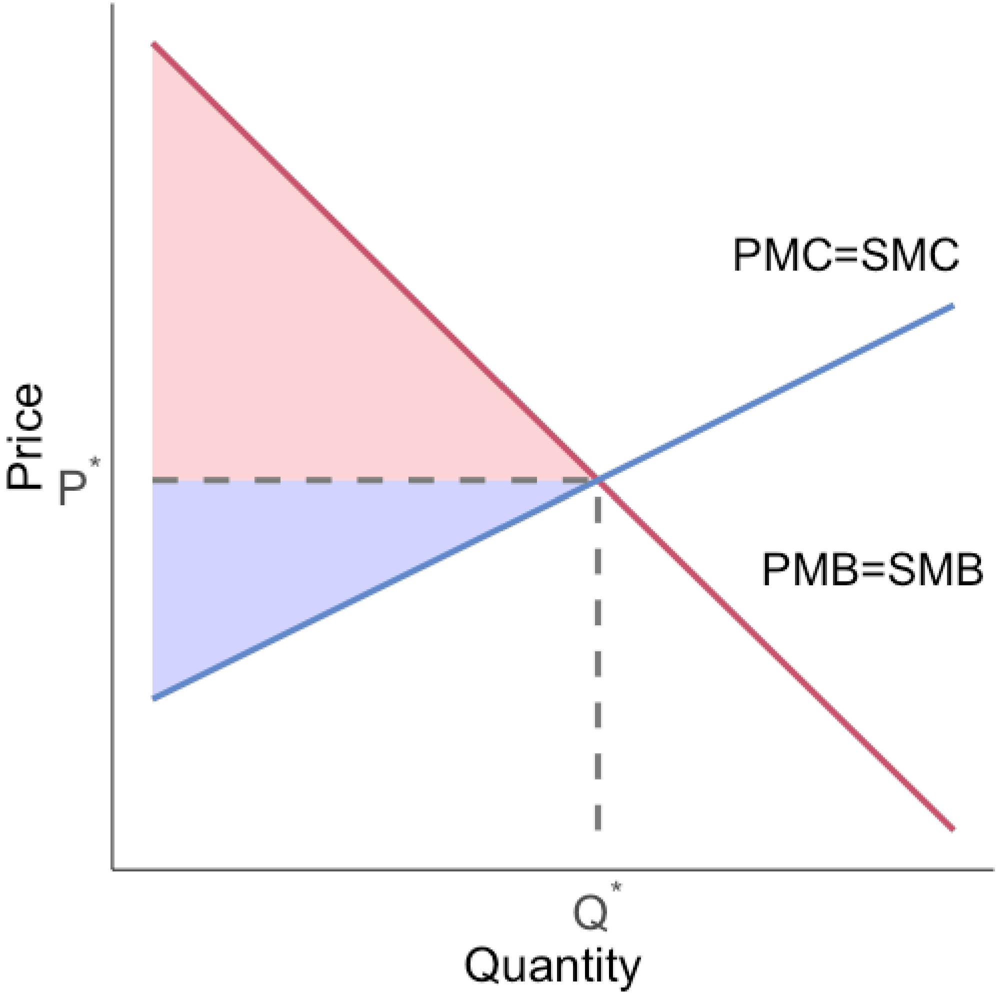
Market equilibrium



The market allocation is **efficient** because SMC = SMB

Why?

Market equilibrium

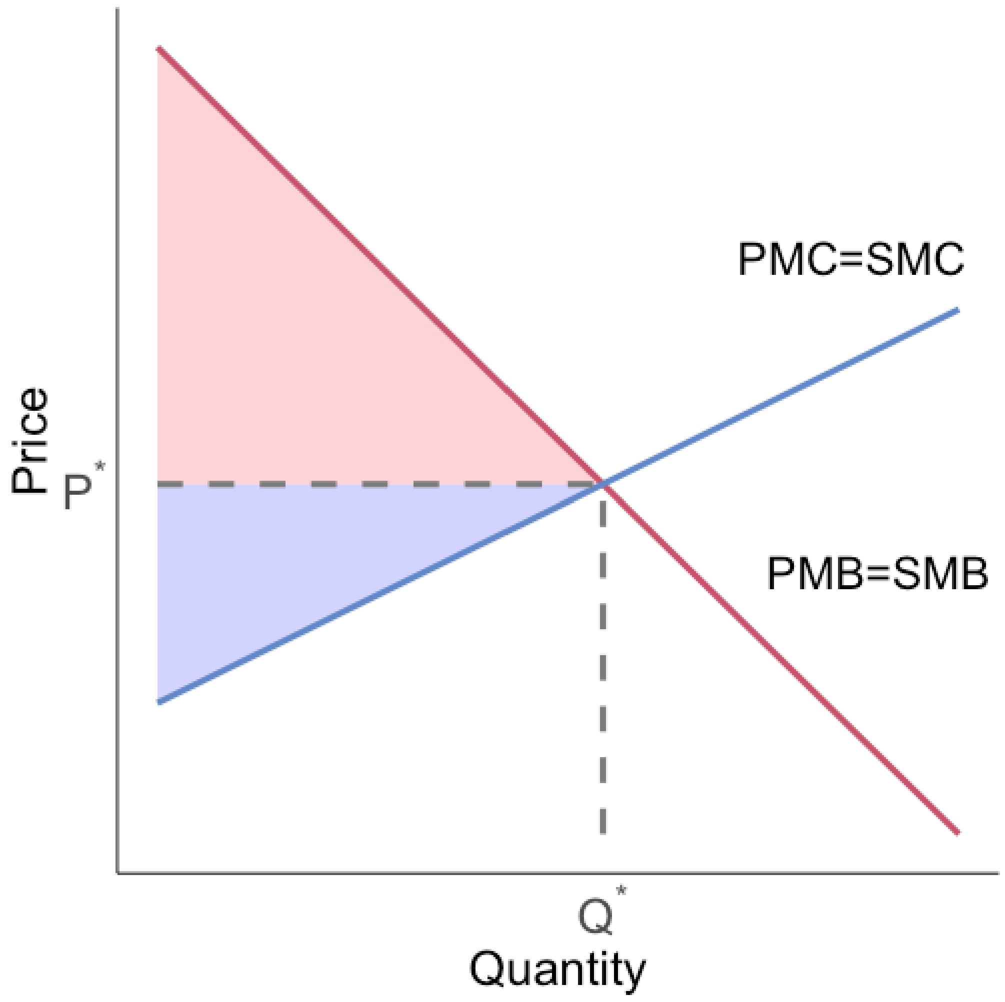


The market allocation is **efficient** because SMC = SMB

Why?

Consider deviating from (P^*, Q^*)

Market equilibrium



Cost of next unit after Q^* > benefit

Benefit of last unit $\geq$ cost of last unit before Q^*

Competitive market allocations are efficient for private goods

Externalities

Externalities

If the world consisted only of competitive markets, private goods, and no third parties, there would be no reason to do anything after Econ 101

That's not the case in the real world

In the real world we have **externalities**

> An externality exists whenever an individual or firm undertakes an action that impacts another individual or firm in an unintended way for which the latter is not compensated (a negative externality), or for which the latter does not pay (a positive externality)

Externalities

What is the problem with externalities for market outcomes and efficiency?

There is not a market for the externality

E.g. at Wegmans or Agway you will not find some important goods on sale:

- Cleaner air outside
- Biodiversity in the Amazon

The central problem is that there are goods that are **not priced**, why is this a problem?

Markets rely on prices to signal the social value of goods

Arizona's water problems share a market-failure structure

Colorado River

- Arizona shares a limited river supply with other states, Tribal Nations, and Mexico
- Each user's withdrawal increases scarcity for other users

Groundwater

- Pumping from a shared aquifer lowers the water table and can cause land subsidence
- Pumpers do not bear all the costs imposed on nearby wells and future users

Both: incomplete property rights + missing prices + high transaction costs

Colorado River rules are being rewritten for 2027

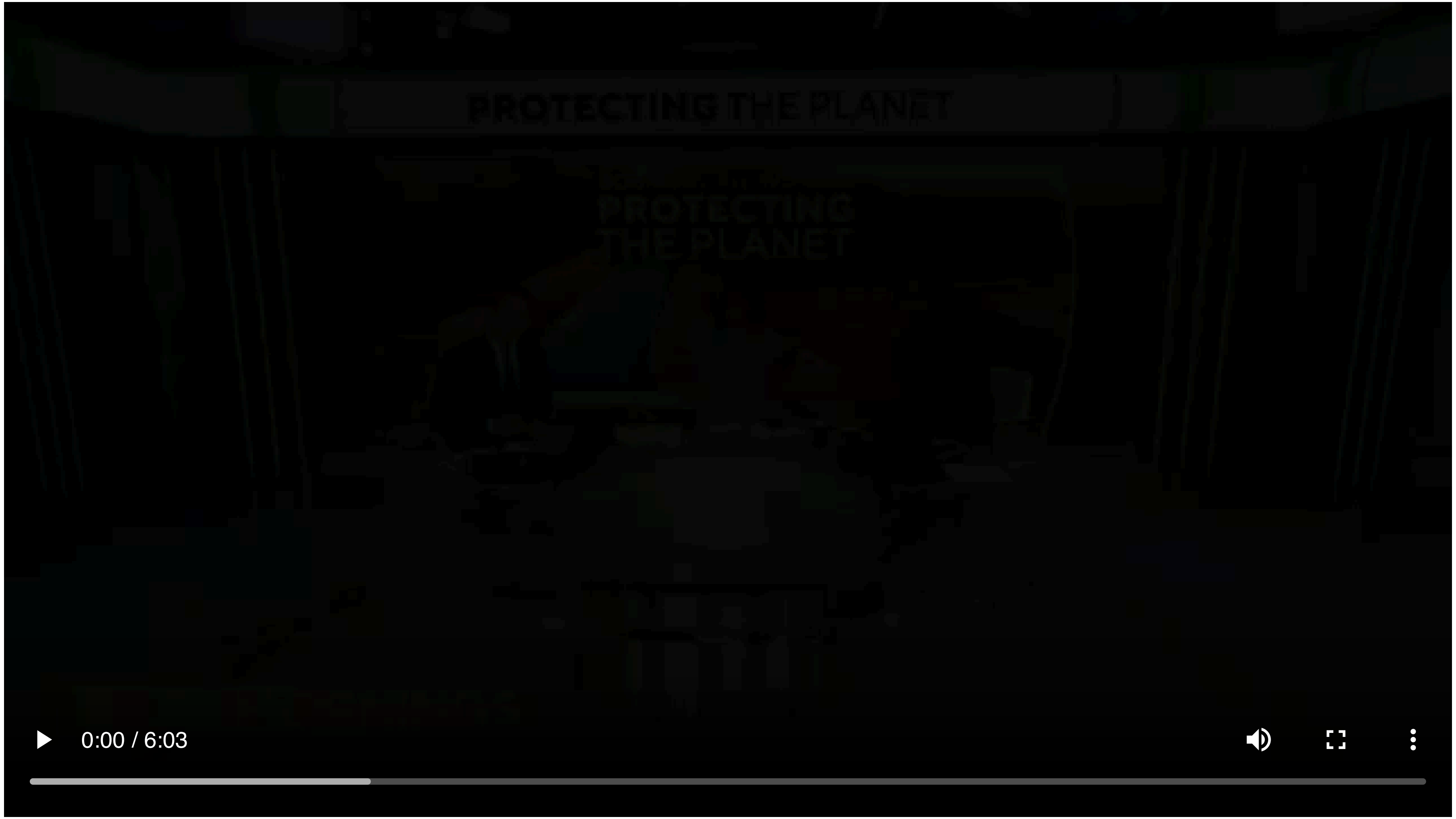
The post-2026 Colorado River regime is not yet settled

As of August 26, 2026:

- Current operating guidelines and drought plans expire at the end of 2026
- The July 31 final environmental impact statement identifies an adaptive framework through 2036
- The Department of the Interior expects final guidelines before October 1

The economic question: who bears shortages when river flows are uncertain?

Saudi farming exposed Arizona groundwater rules



Arizona has begun regulating Ranegras groundwater

- Annual withdrawals have exceeded natural recharge by roughly **900%**
- One monitored well has fallen more than **240 feet** since the 1980s, and groundwater loss has caused land subsidence
- Arizona designated Ranegras Plain as its eighth Active Management Area in January 2026
- The new rules restrict irrigation on new acreage and require large users to measure and report pumping

The change: an open-access aquifer is becoming a managed common-pool resource

Groundwater scarcity is also moving through the courts

- Arizona sued Fondomonte in December 2024, alleging that excessive pumping threatened nearby communities and infrastructure
- In February 2026, the attorney general argued that the new Active Management Area complements rather than replaces the lawsuit
- The state is seeking remedies targeted at the alleged harm, while the Active Management Area regulates pumping throughout the basin

Same externality, different tools: basin-wide regulation and liability for alleged harm

Groundwater enforcement now uses settlements as well as lawsuits

Externalities: several classifications

We can classify externalities in a few ways:

Production externalities: generated by a firm in the process of producing some output (e.g. pollution, innovation)

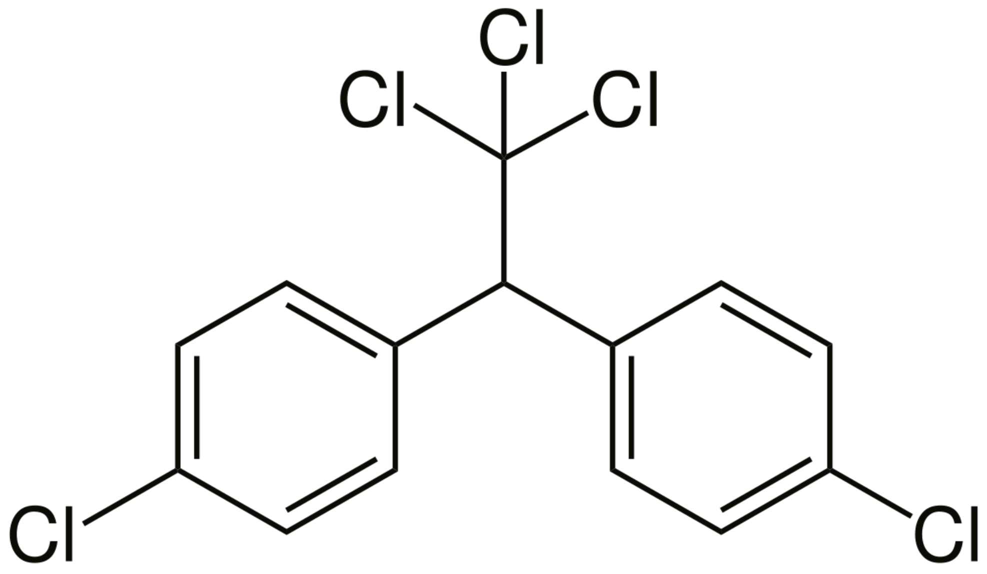
Consumption externalities: generated by an individual in the process of consuming an output (e.g. congestion, vaccination)

Negative externalities: imposes external **costs** (e.g. pollution)

Positive externalities: imposes external **benefits** (e.g. vaccination)

Negative externalities: what is this?

Negative externalities: DDT, shockingly bad for you

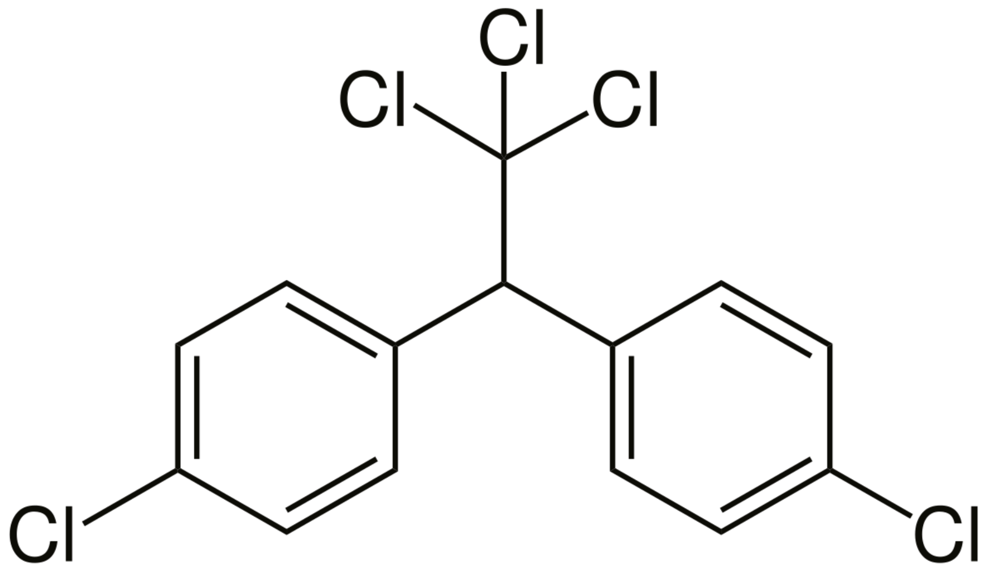


DDT is a chemical that was widely used as an insecticide in the early-to-mid 1900s

Widely used to eradicate Typhus and Malaria

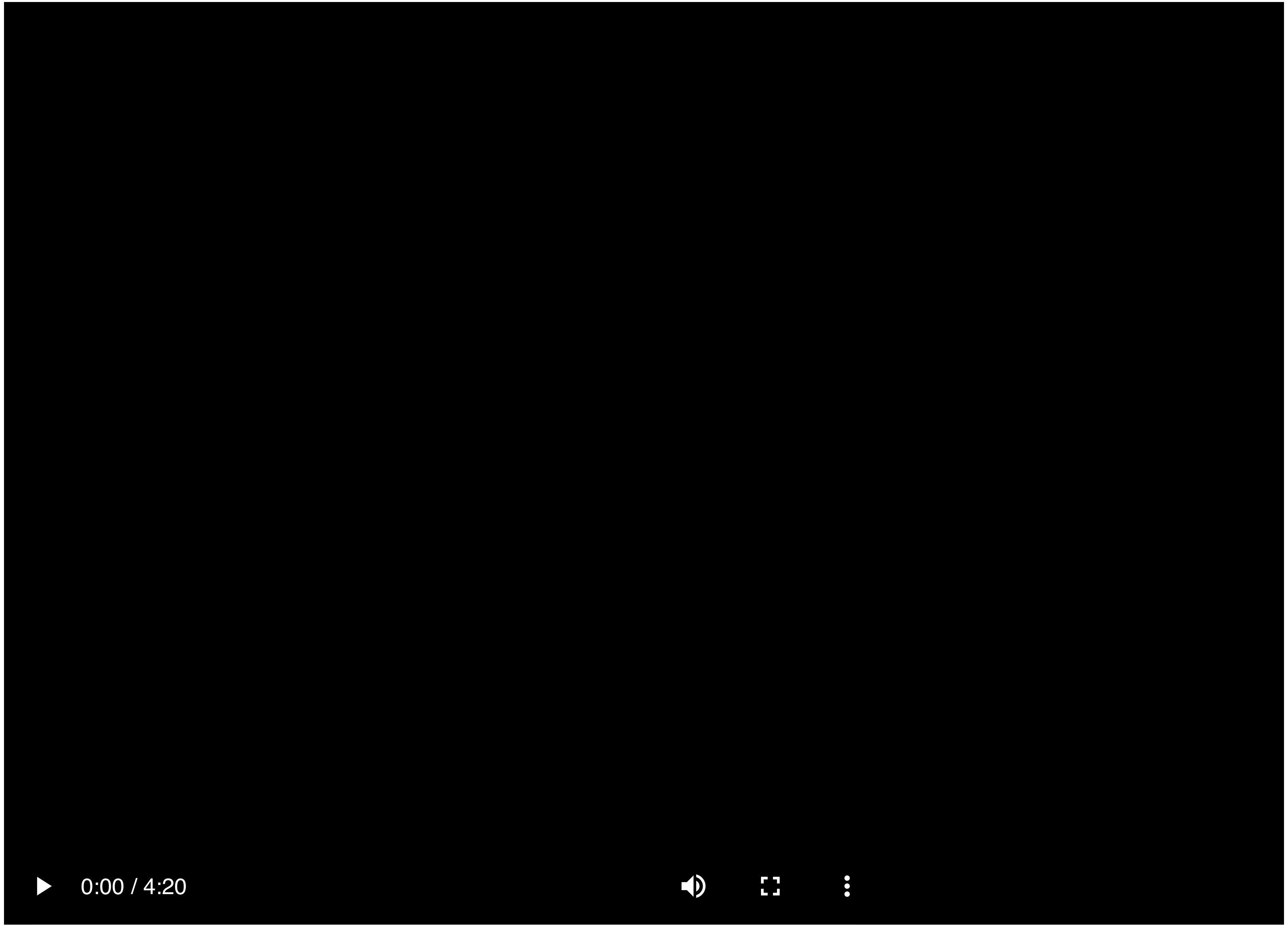
Used to treat lice

Negative externalities: DDT, gives you cancer

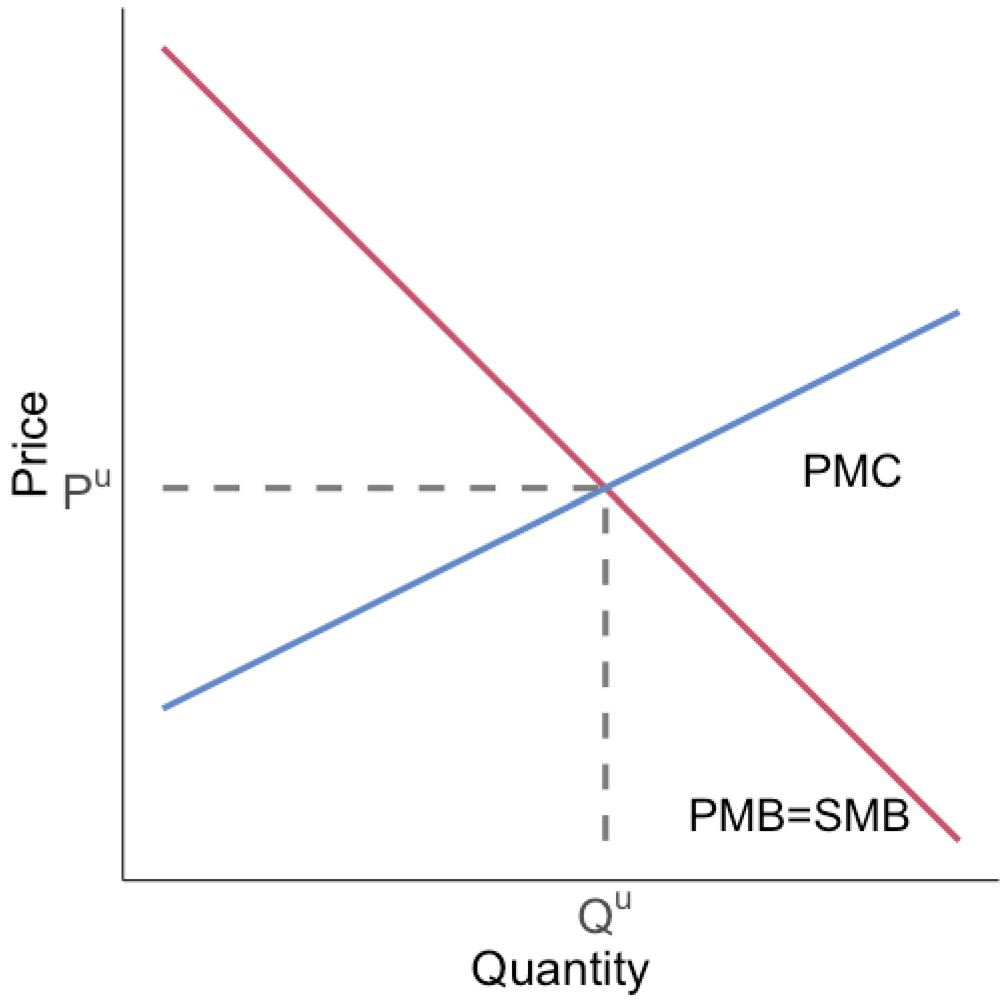


A relationship between DDT exposure and reproductive effects in humans is suspected, based on studies in animals. In addition, some animals exposed to DDT in studies developed liver tumors. As a result, today, DDT is classified as a probable human

The birth of the environmental movement



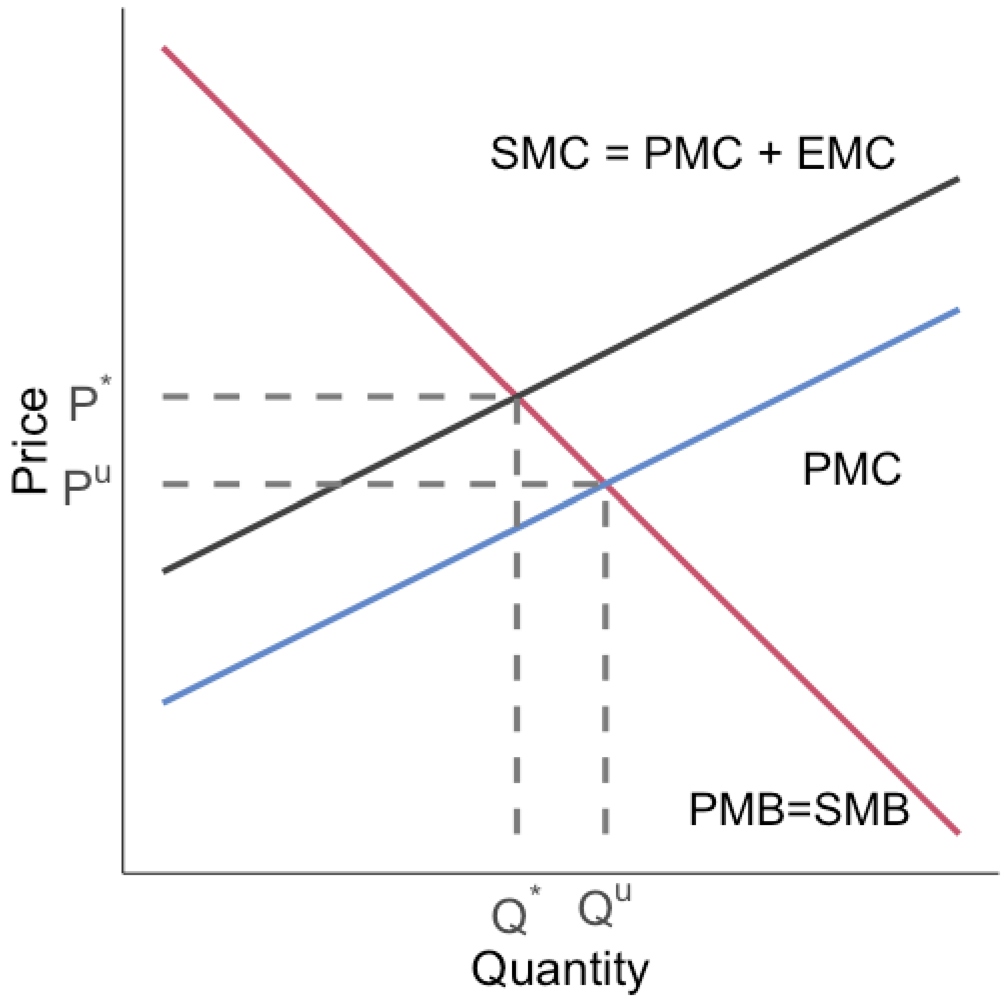
Negative externalities: graphical



Social marginal cost (SMC) is the sum of private marginal cost (PMC) and the external marginal cost (EMC)

Where is the SMC?

Negative externalities: graphical

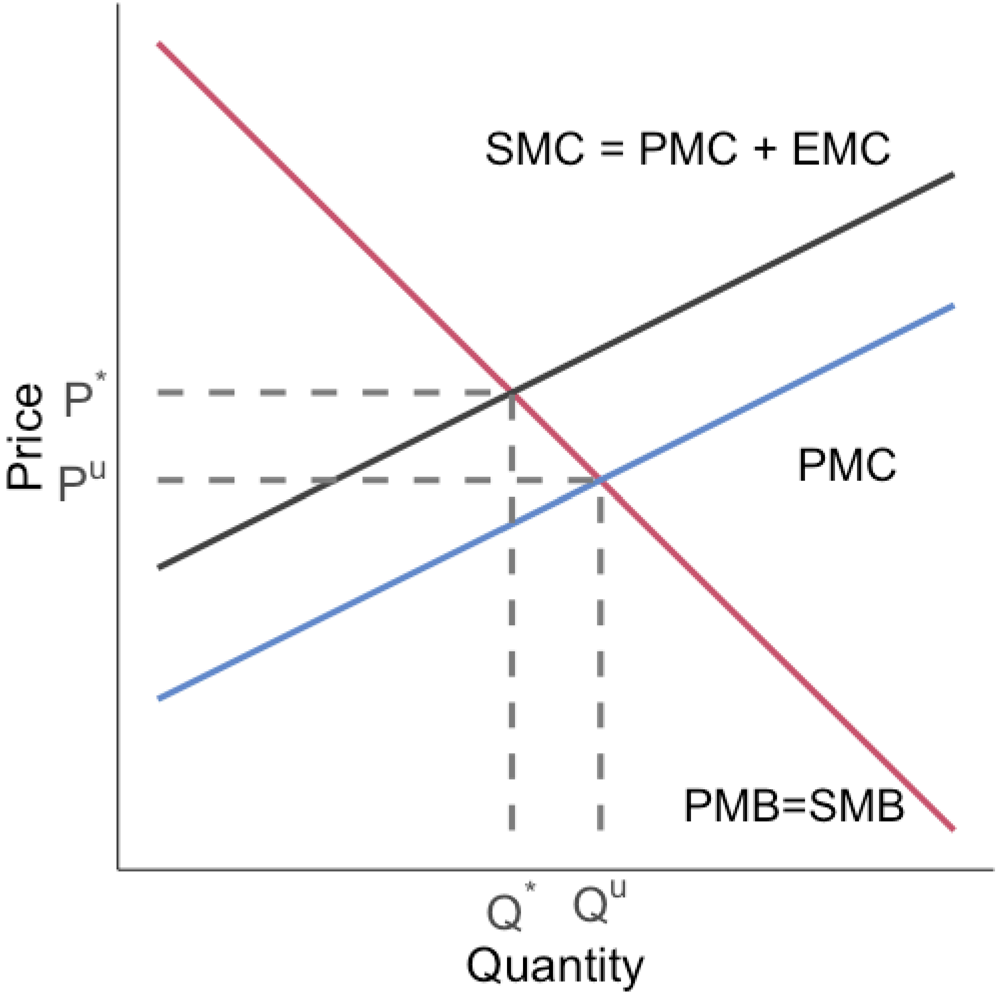


Social marginal cost (SMC) is the sum of private marginal cost (PMC) and the external marginal cost (EMC)

The PMC curve only reflects the **private costs** of making the DDT

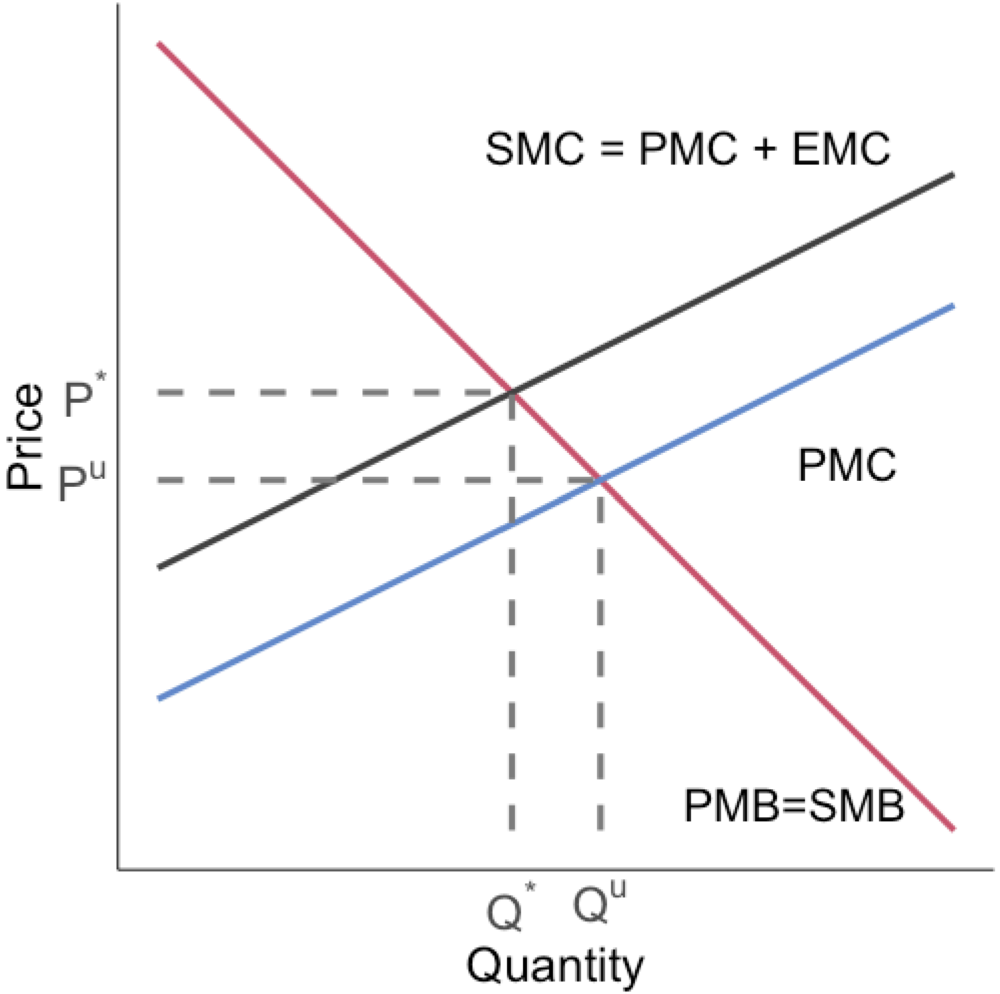
It does not account for the external health and wildlife costs

Negative externalities: graphical



Adding the private and external marginal costs together gives us the SMC, what we care about from the social planner or regulator's perspective

Negative externalities: graphical



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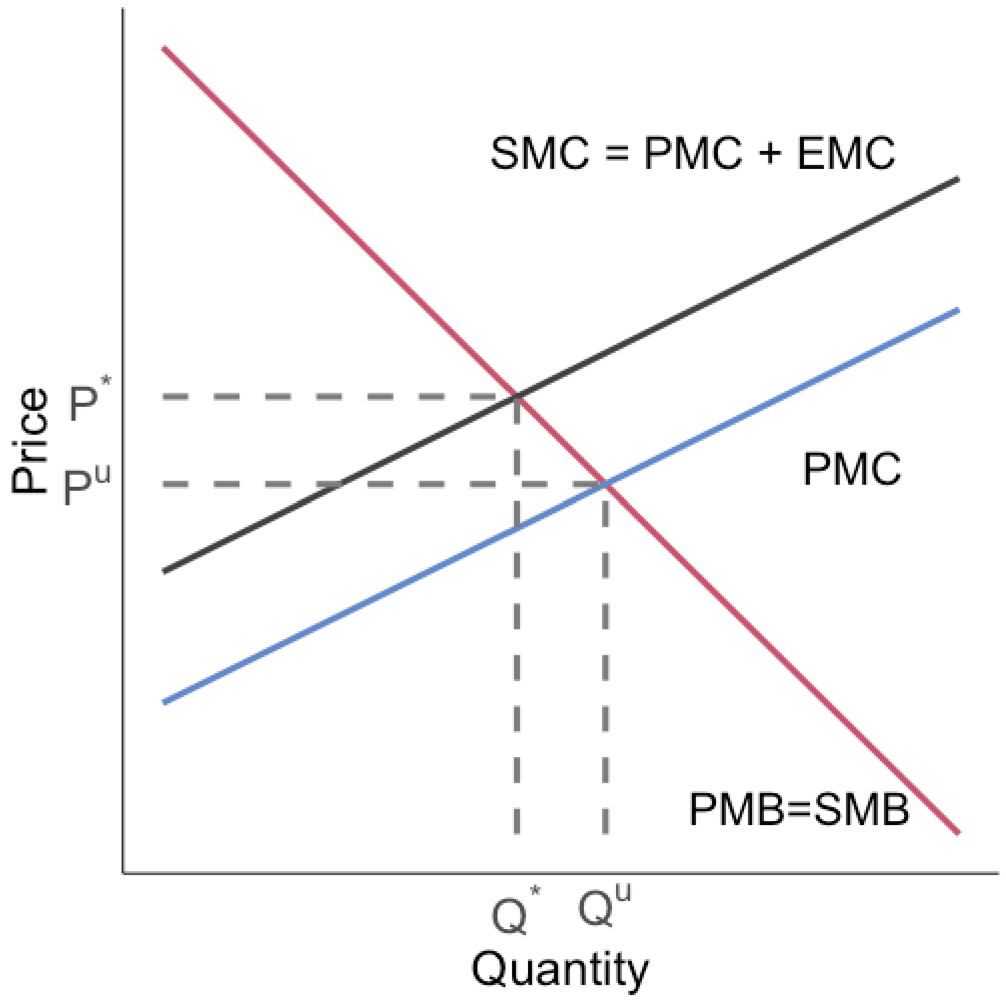
The unregulated market gives us (P^u, Q^u) as an outcome when we want (P^*, Q^*)

Negative externalities: graphical



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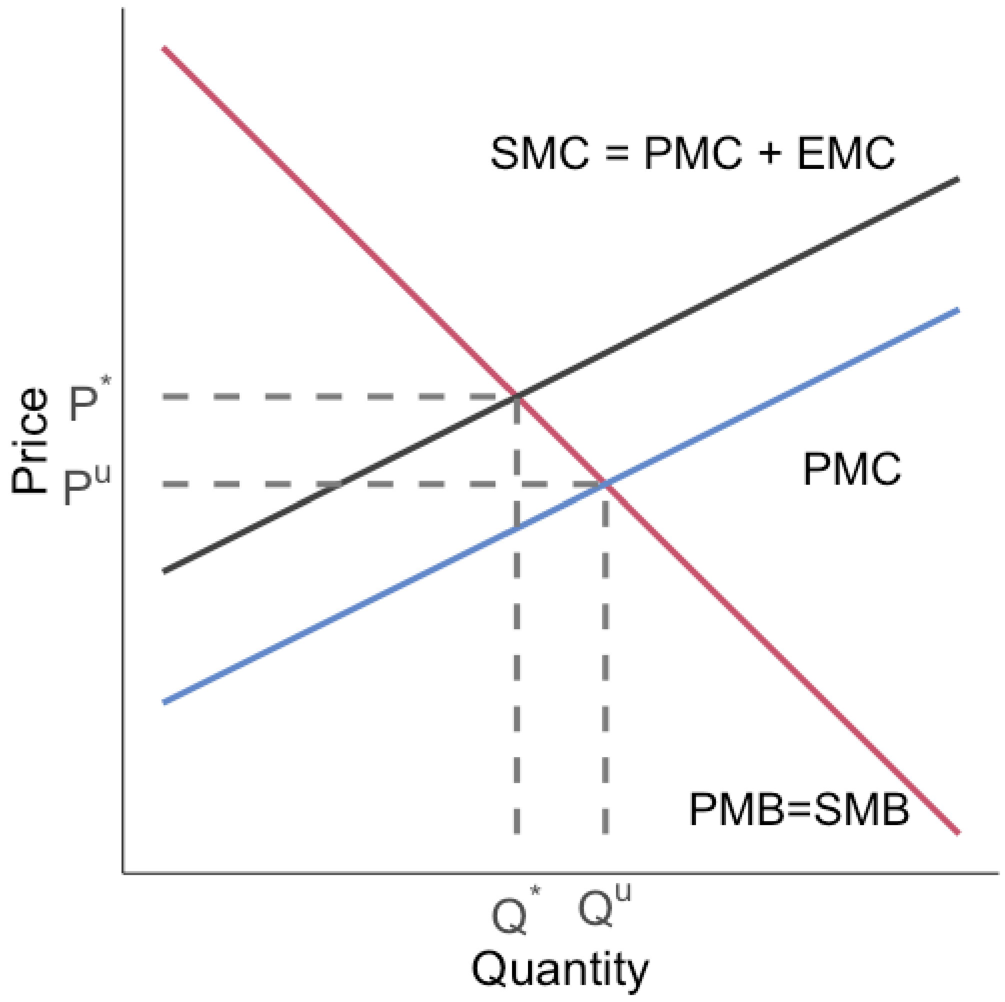
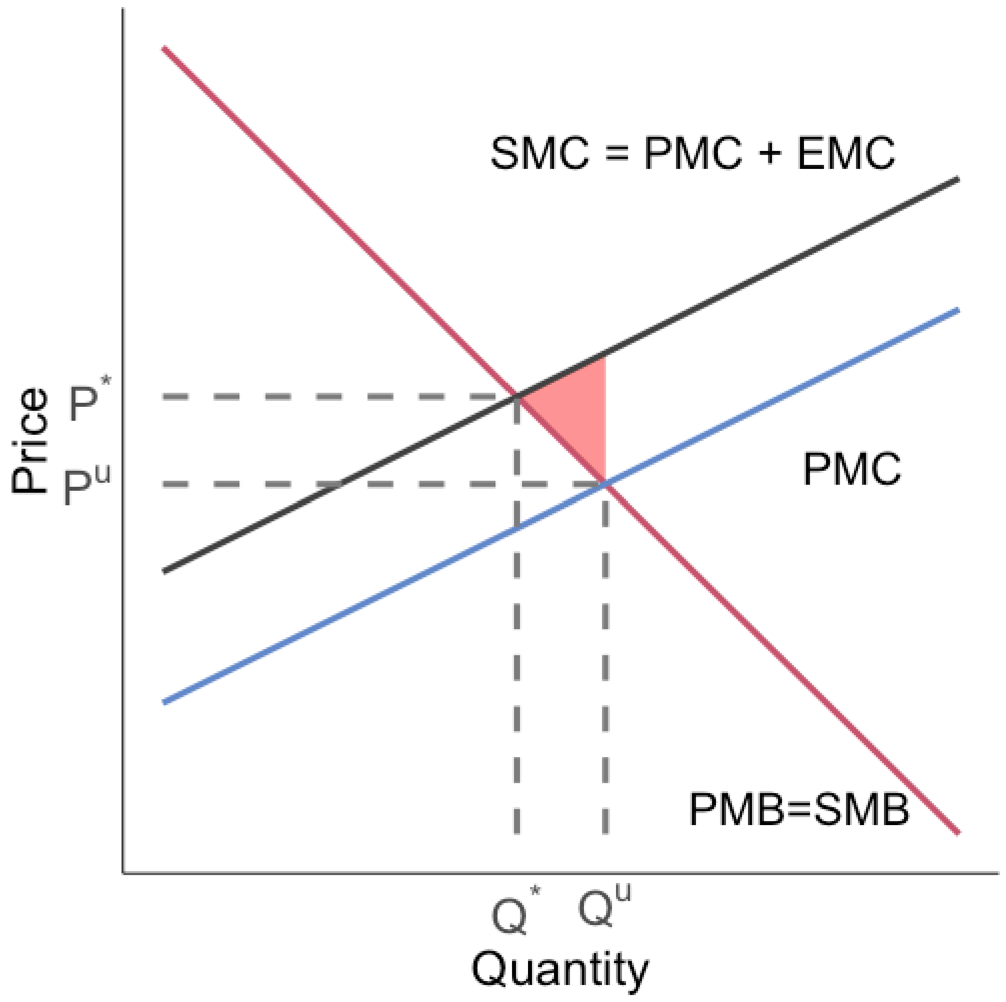
What's the social cost of this market failure?

Negative externalities: graphical



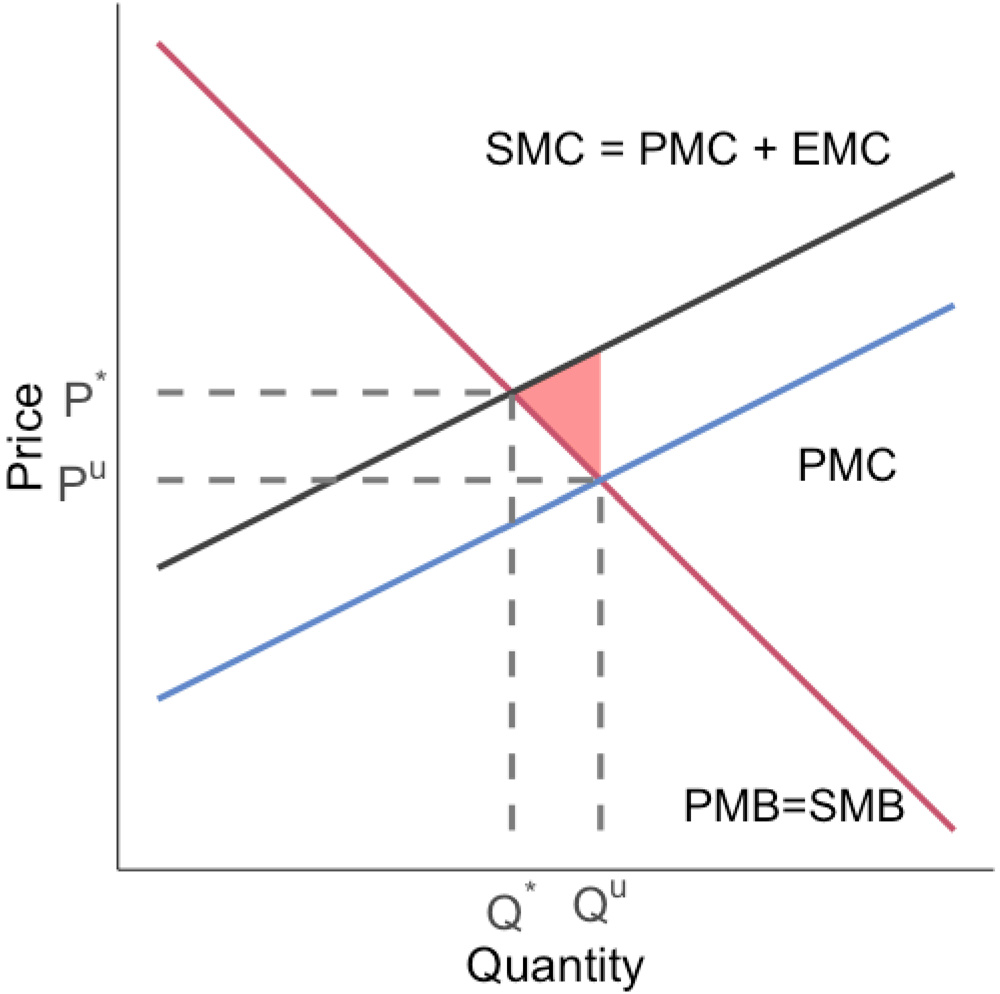
Negative externalities generate deadweight loss equal to...

Negative externalities: graphical



Negative externalities generate deadweight loss equal to the **red** area

Negative externalities: graphical

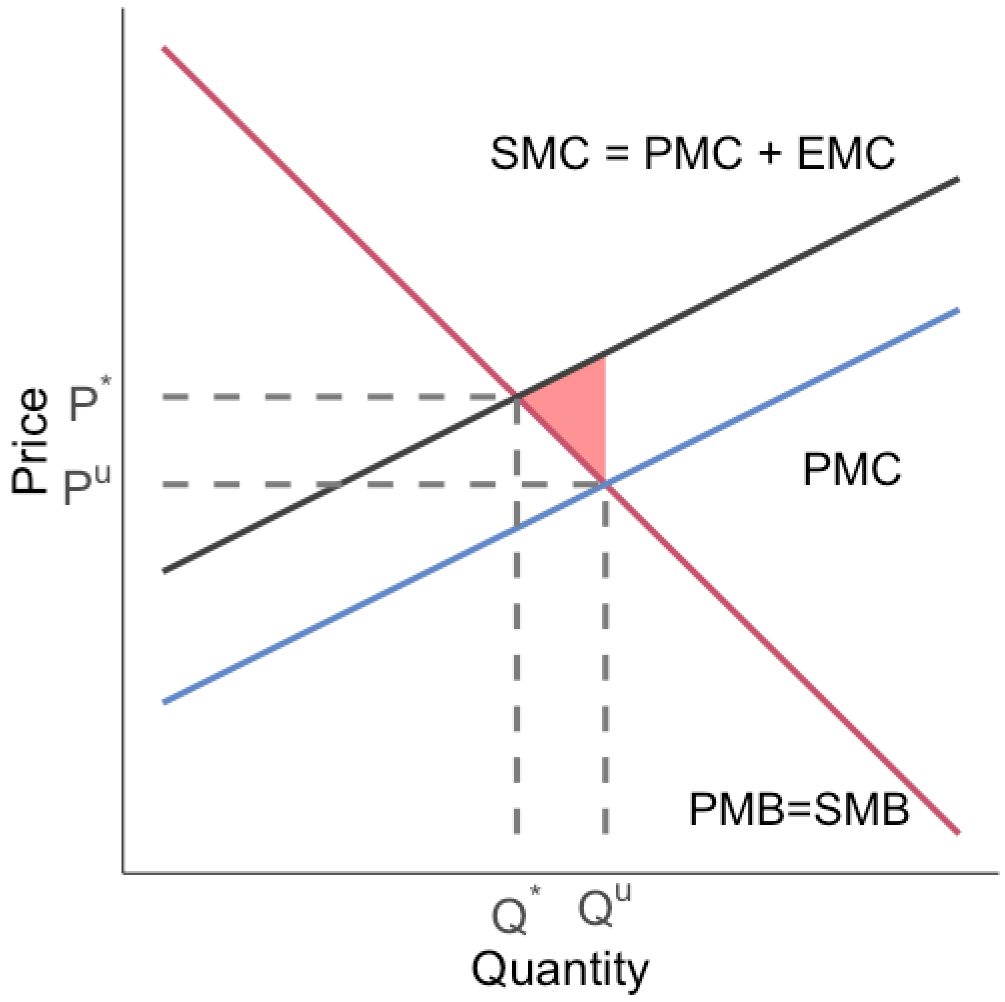


Negative externalities generate deadweight loss equal to the **red** area

This is the difference in SMC and SMB for units bought/sold where SMC > SMB:

Total SMC - SMB from Q^* to Q^u

Negative externalities: graphical

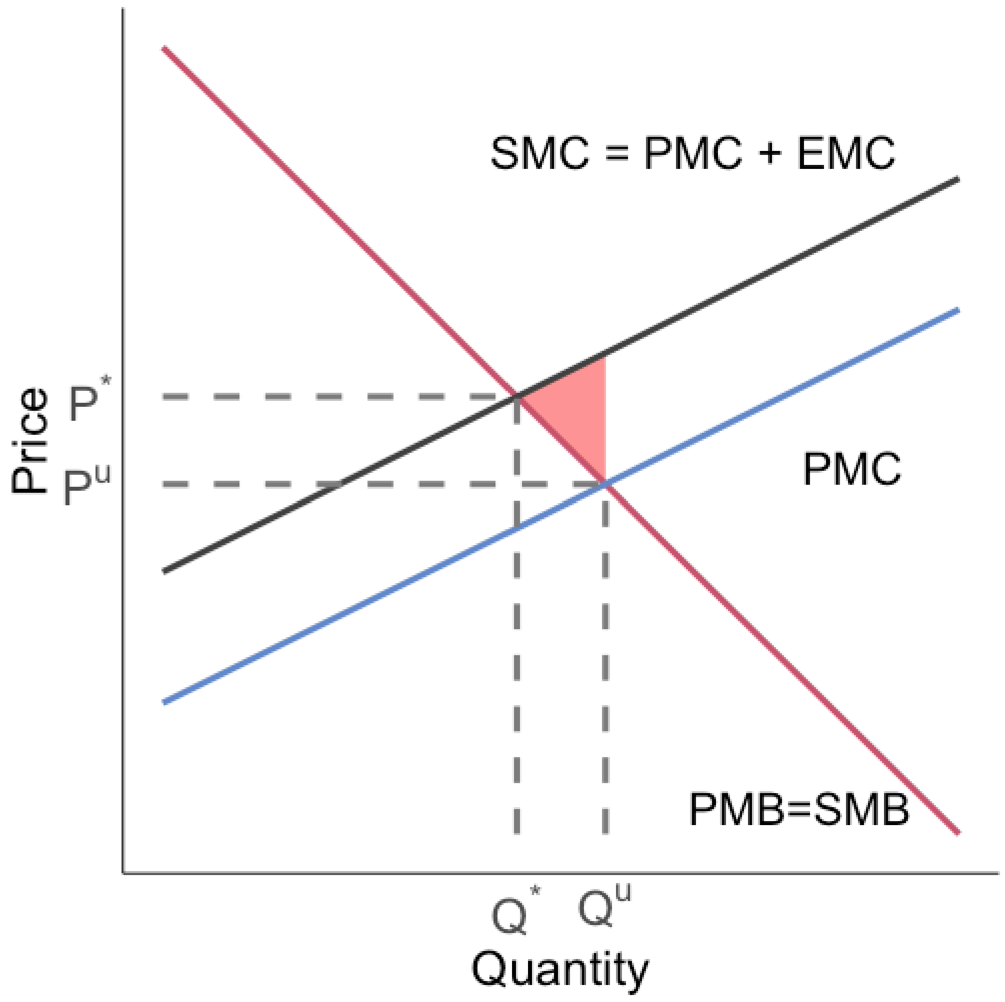


Negative externalities generate deadweight loss equal to the **red** area

This is the difference in SMC and SMB for units bought/sold where SMC > SMB

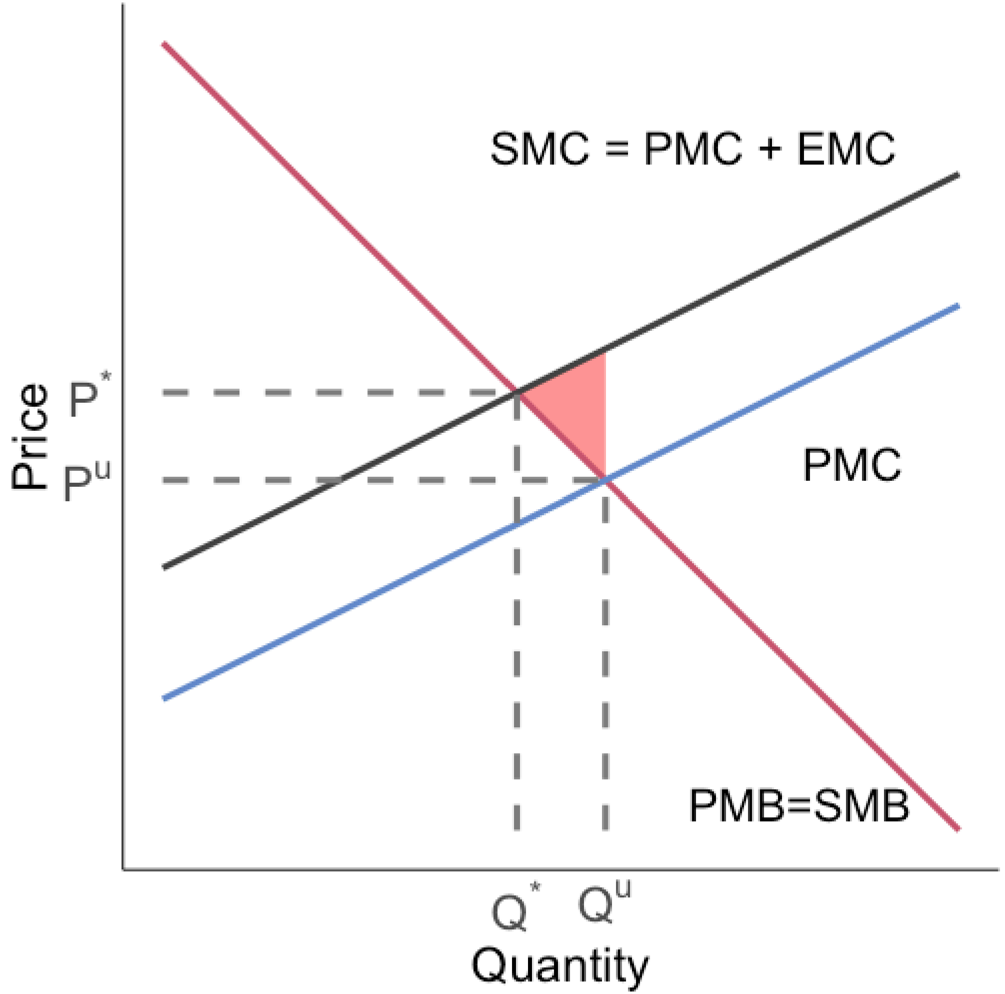
This is the loss to society caused by the externality in the unregulated private market

Negative externalities: graphical



Key takeaway:

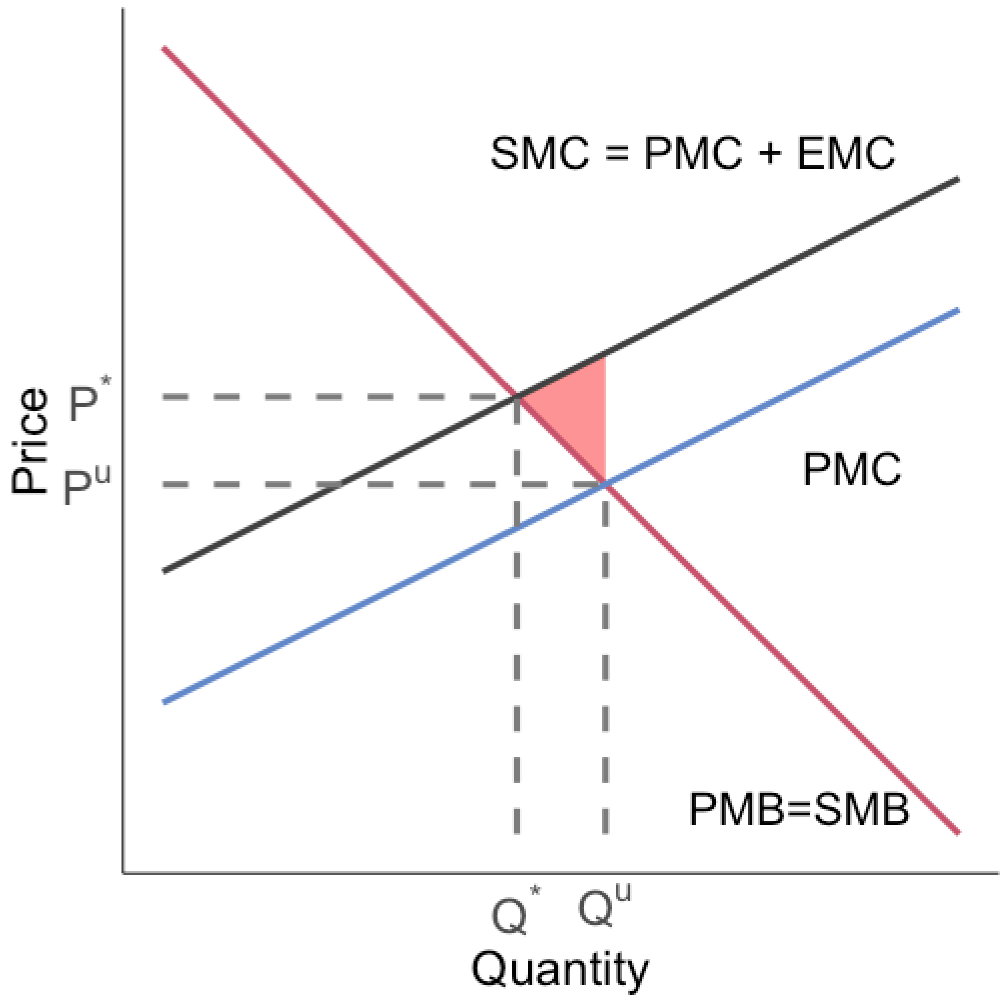
Negative externalities: graphical



Key takeaway:

The private market produces too much DDT

Negative externalities: graphical



Key takeaway:

The private market produces too much DDT

The private actors are not accounting for the **external costs** they are imposing on people who are not in the DDT transaction (e.g. third parties whose health is affected)

Estimating marginal damages with EZ-Pass

Positive externalities

Visualizing herd immunity

If enough people have immunity, the virus is less likely to spread because the few who aren't immune are less likely to come in contact with someone who is infected.

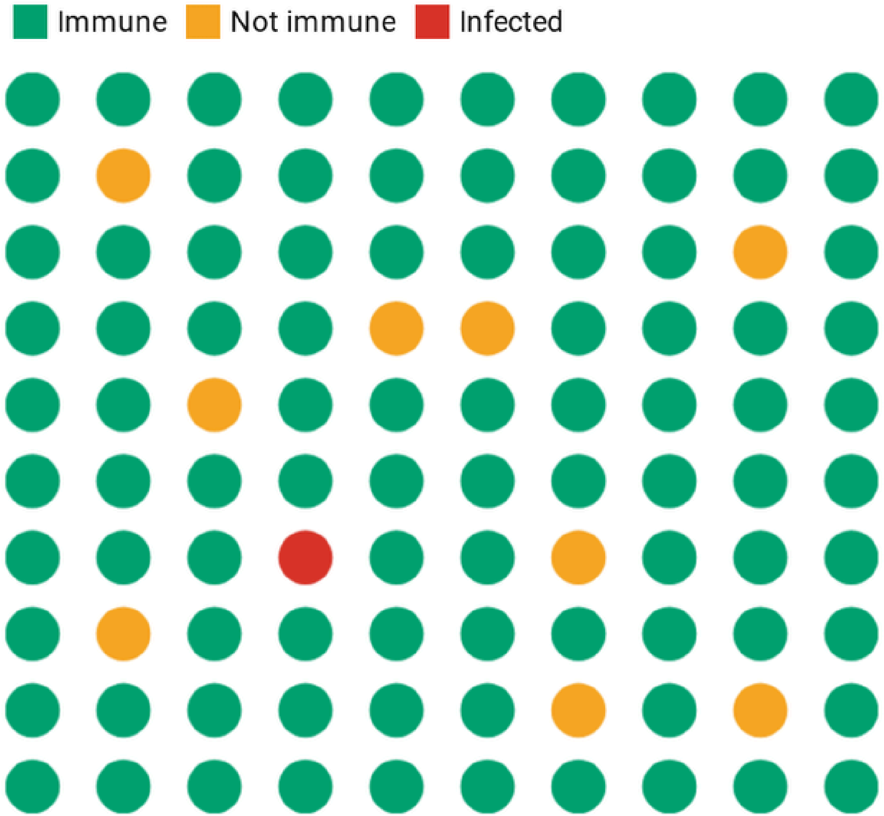


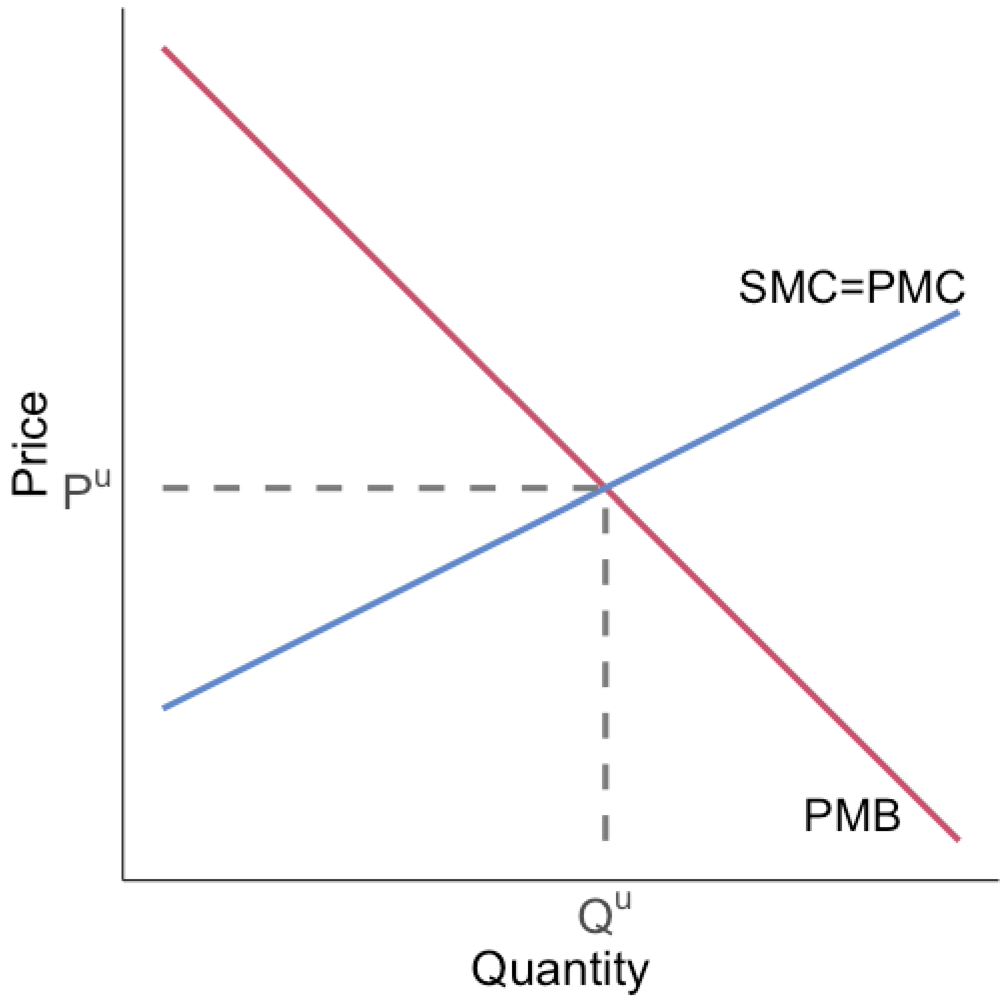
Chart: The Conversation, CC-BY-ND • Source: Sara Krehbiel

Positive externalities

Vaccines and masks are examples of goods with positive externalities

Getting a vaccine or wearing a mask can benefit people who are not involved in that decision

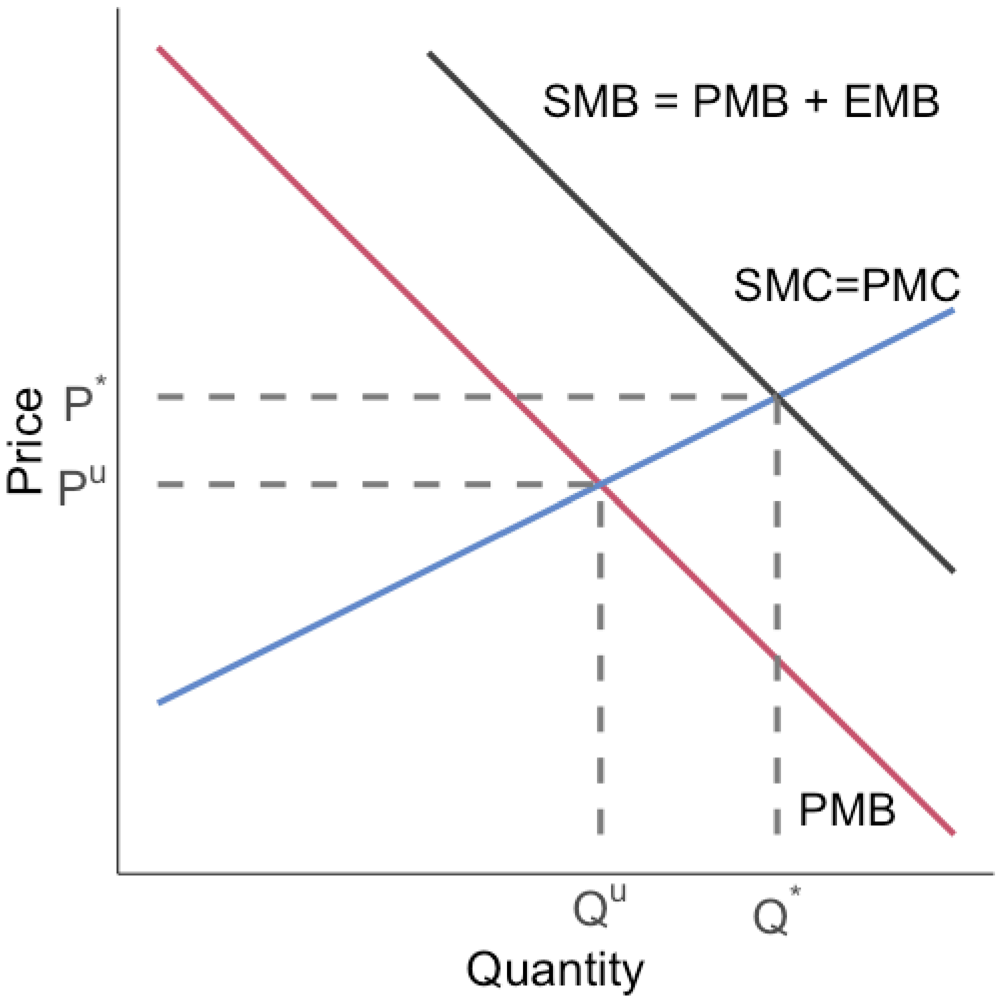
Positive externalities: graphical



Social marginal benefit (SMB) is the sum of private marginal benefit (PMB) and the external marginal benefit (EMB)

Where does the SMB curve lie?

Positive externalities: graphical



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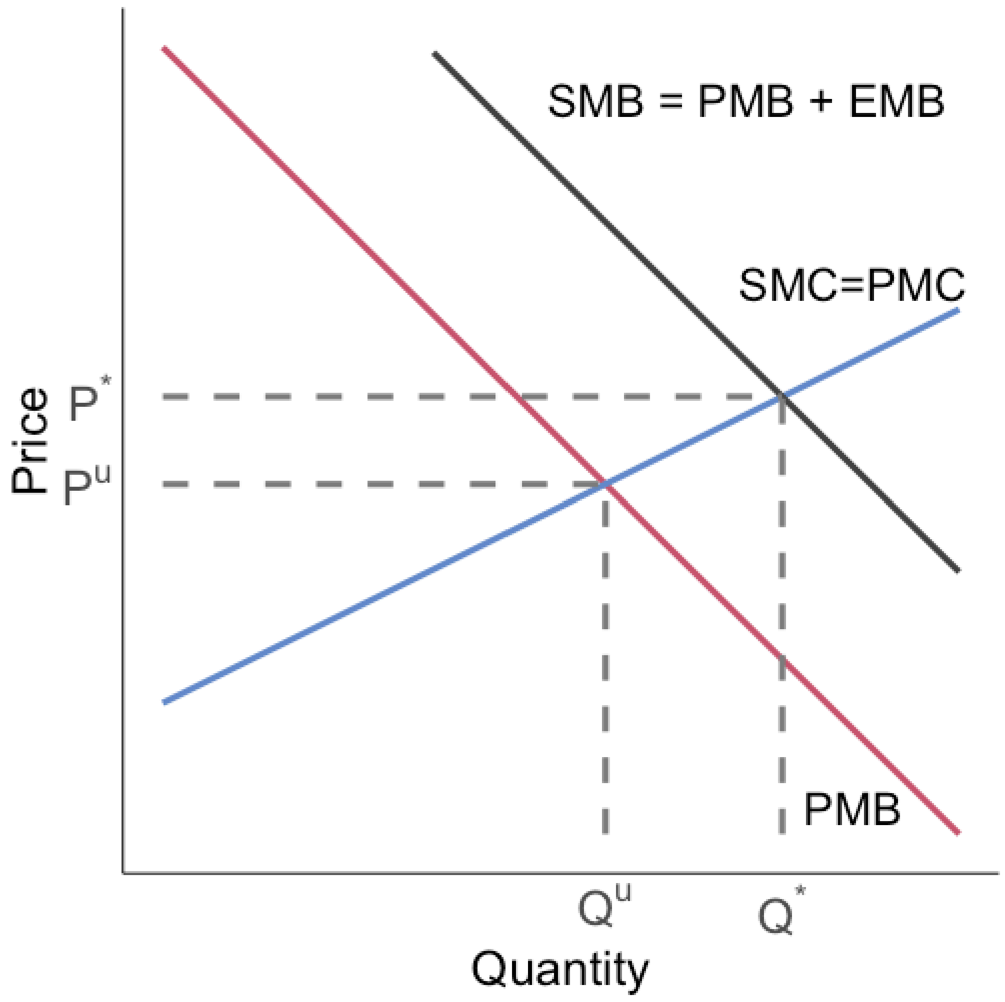

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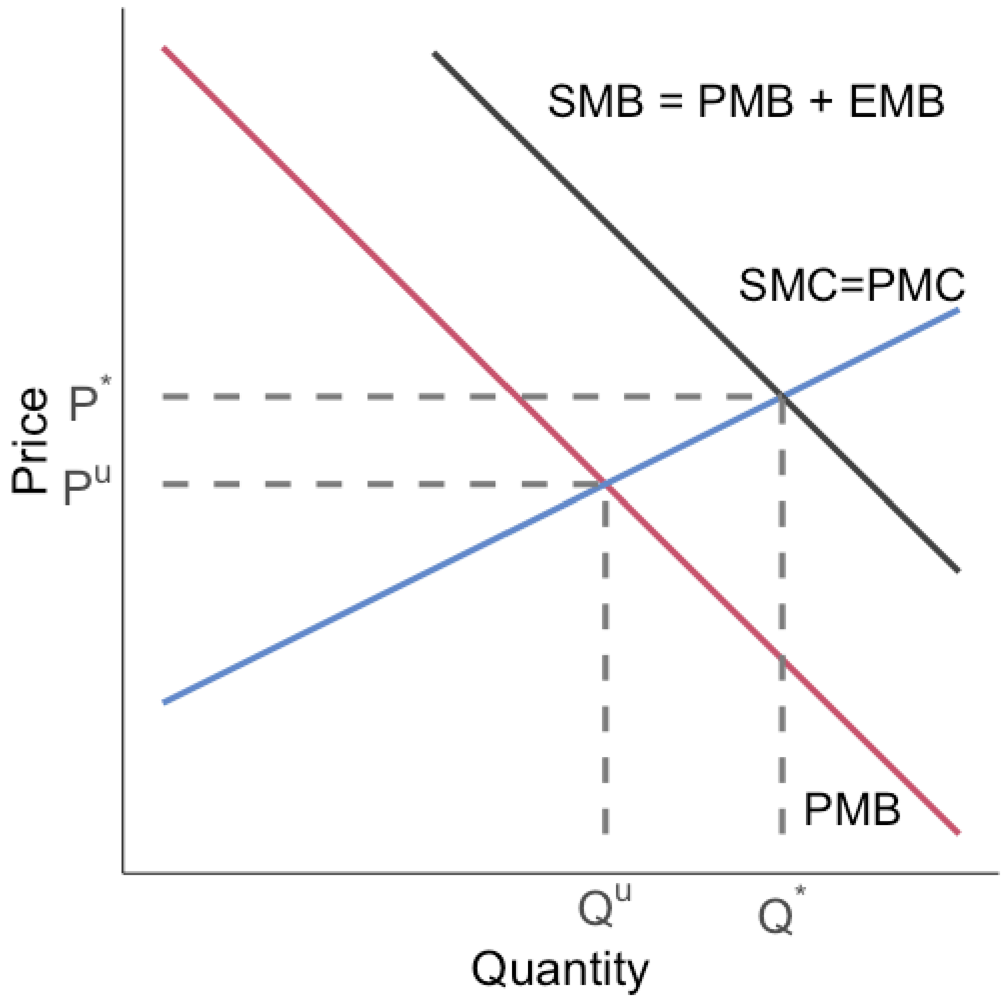

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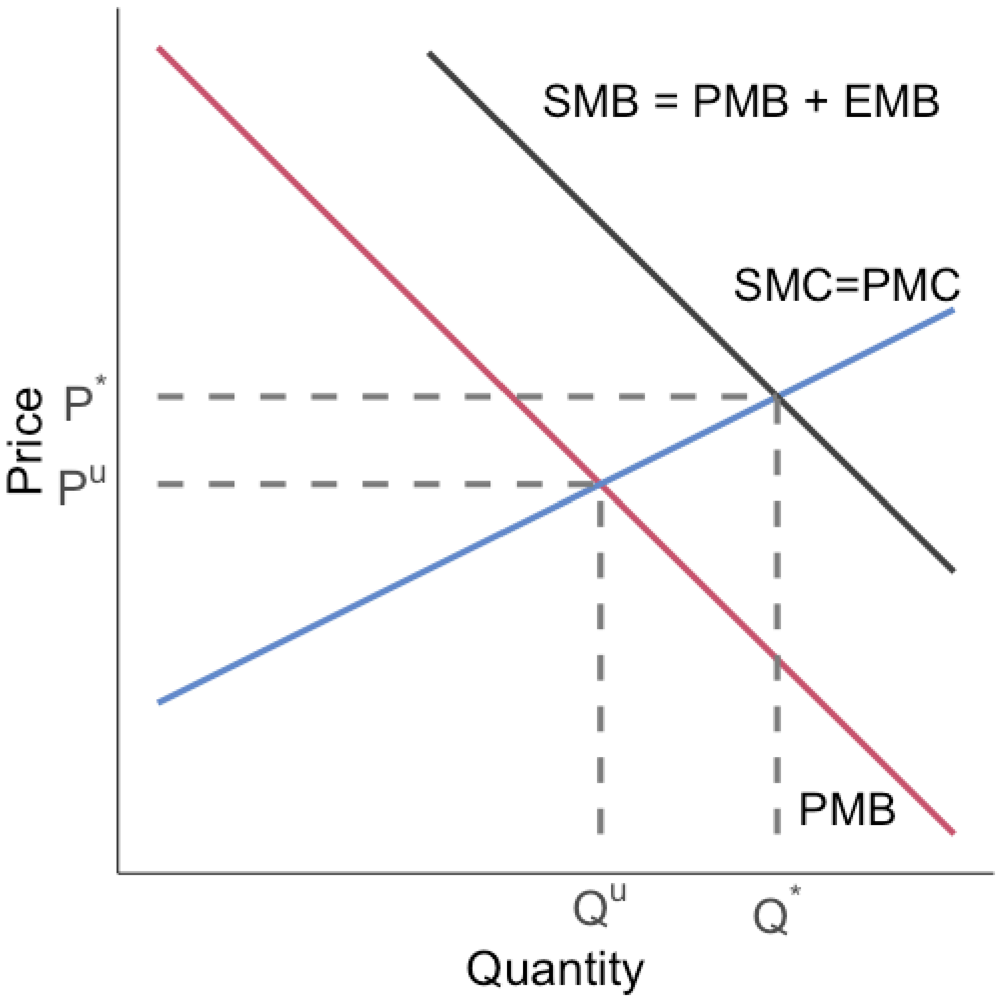
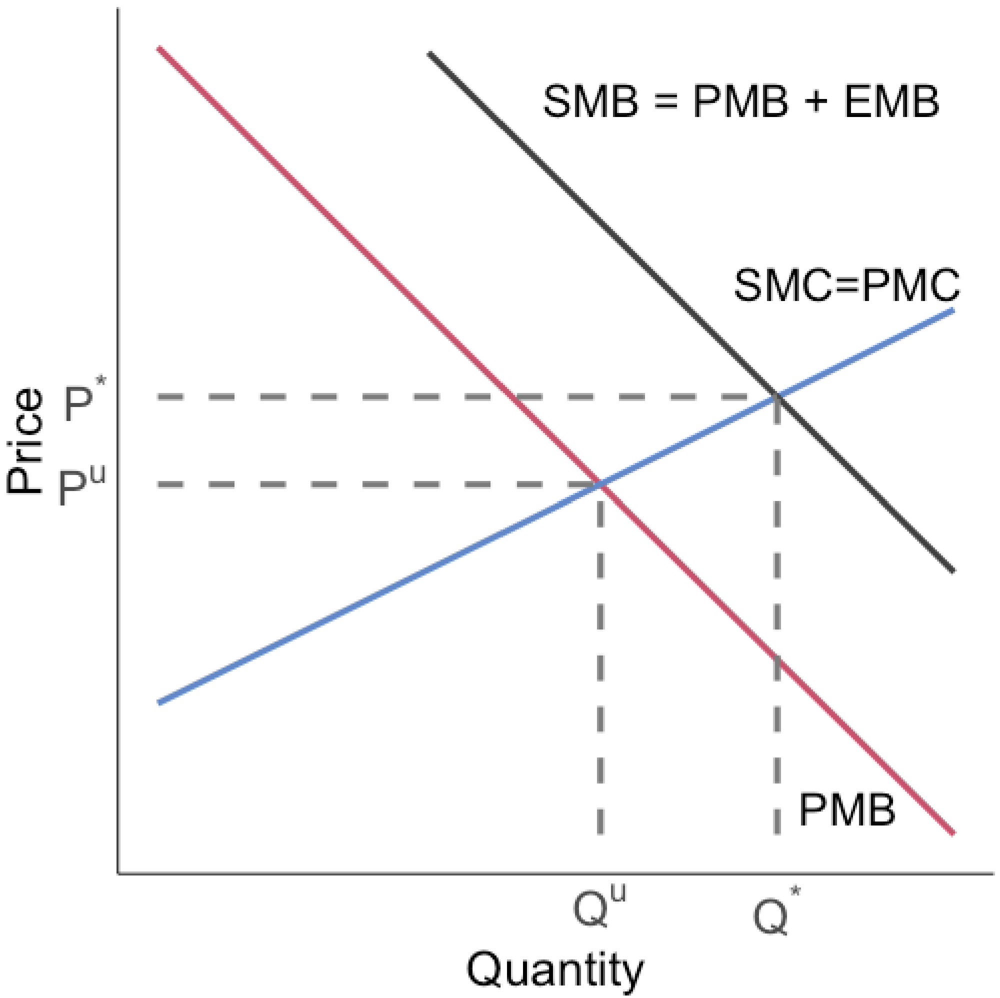
It does not account for the external herd immunity benefits

Positive externalities: graphical



Adding the private and external marginal benefits together gives us the SMB, what we care about from the social planner or regulator's perspective

Positive externalities: graphical



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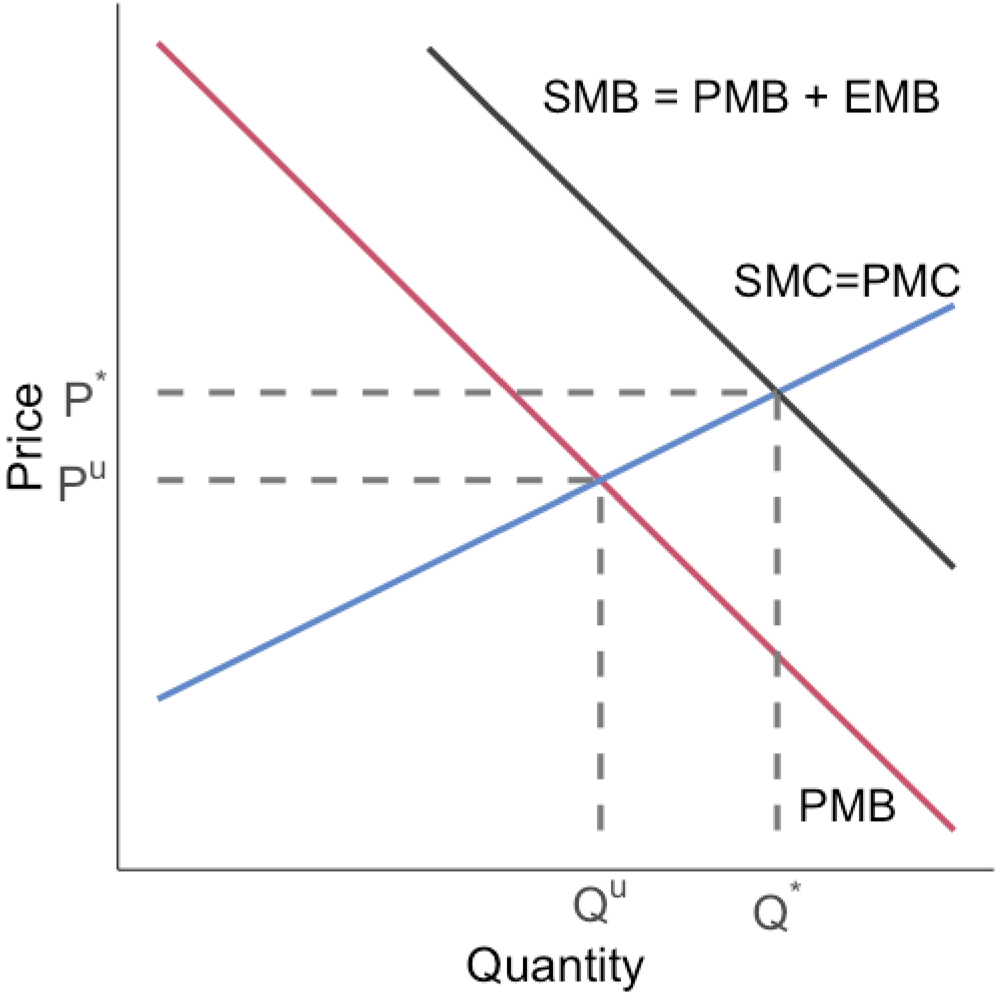

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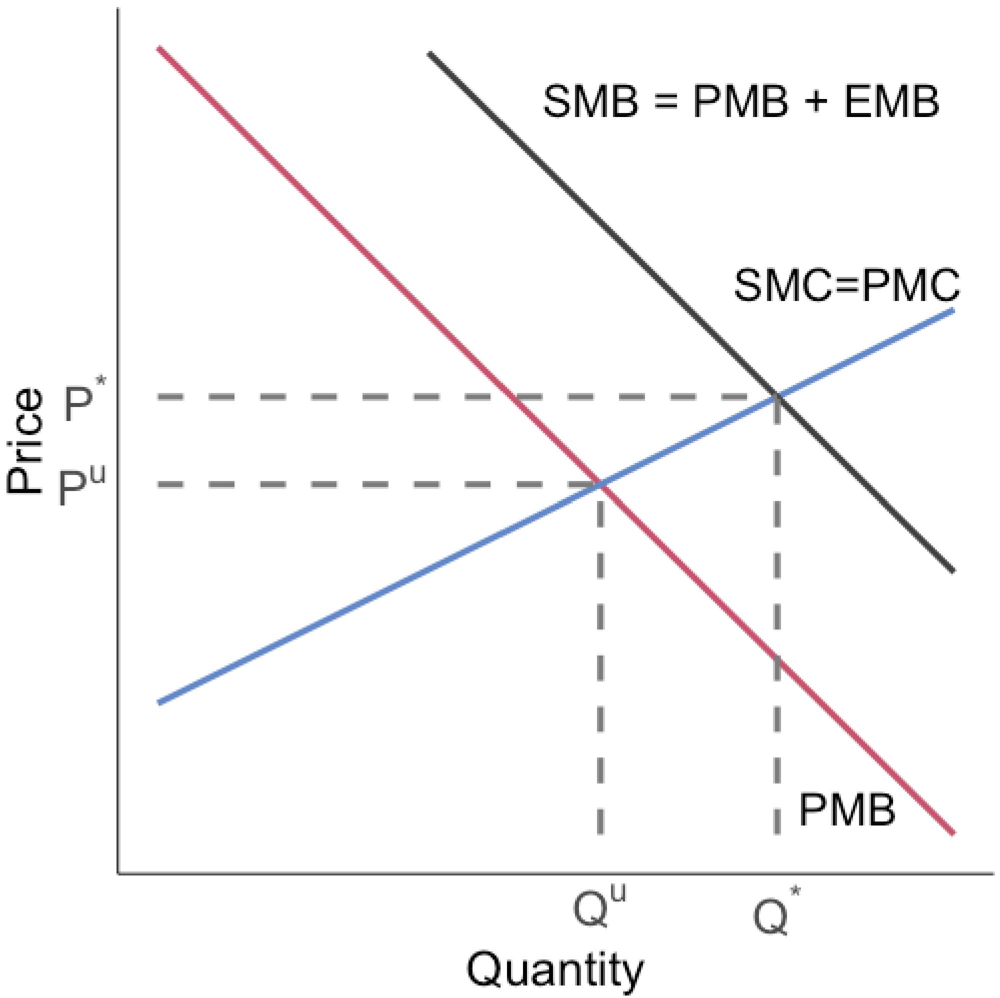
Positive externalities: graphical



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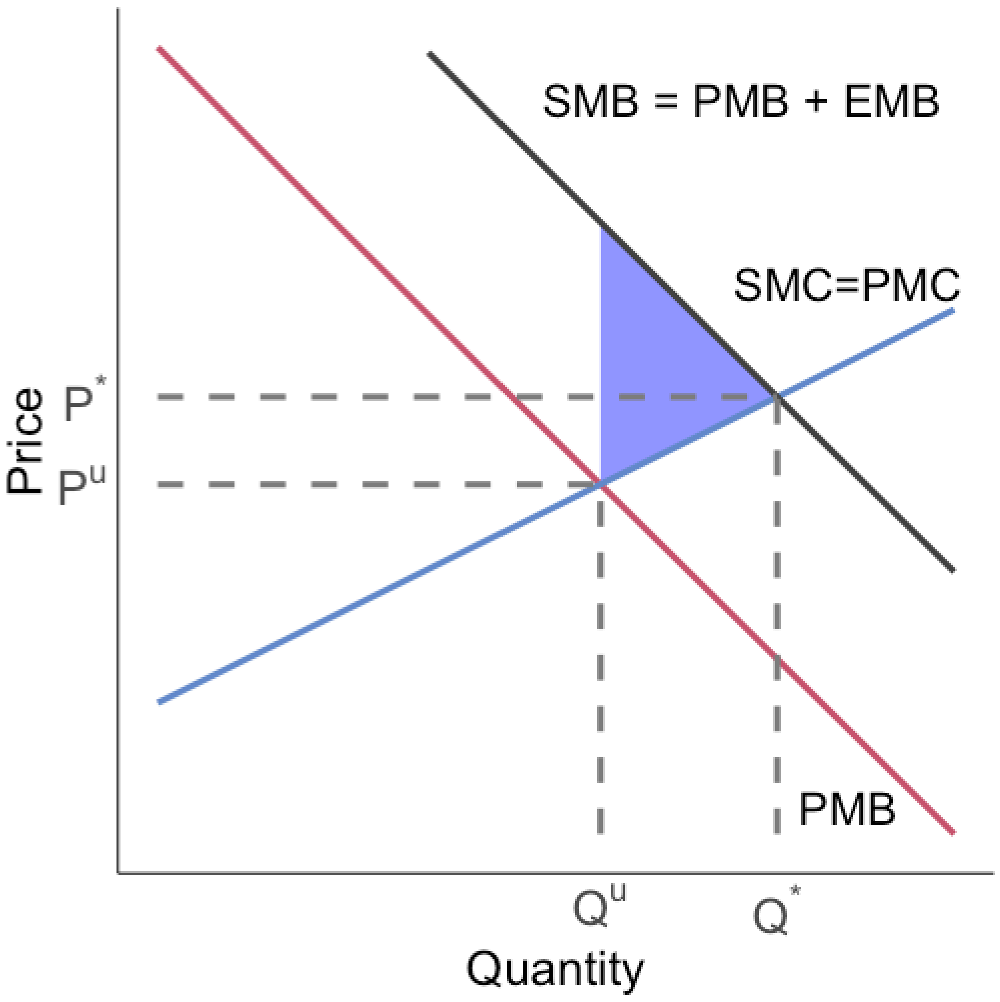
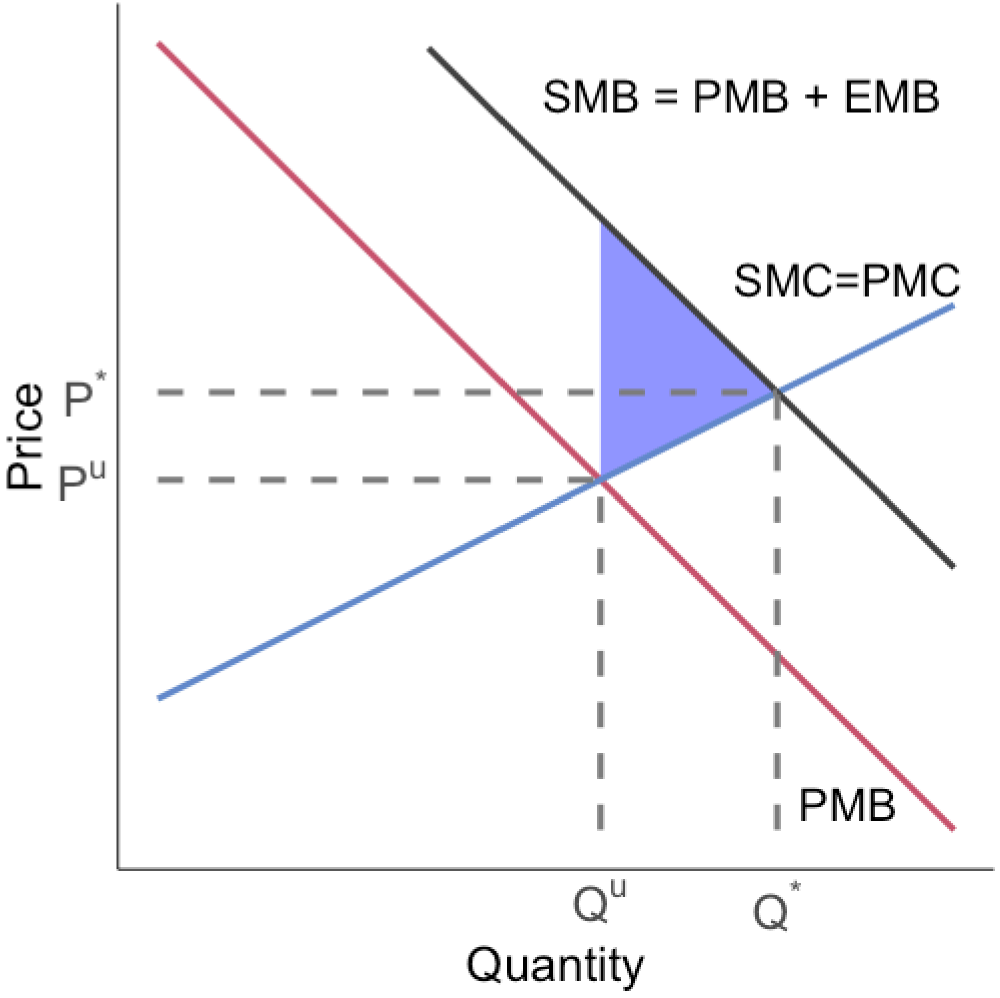
Positive externalities: graphical



Positive externalities generate deadweight loss equal to the **blue** area

Positive externalities: graphical



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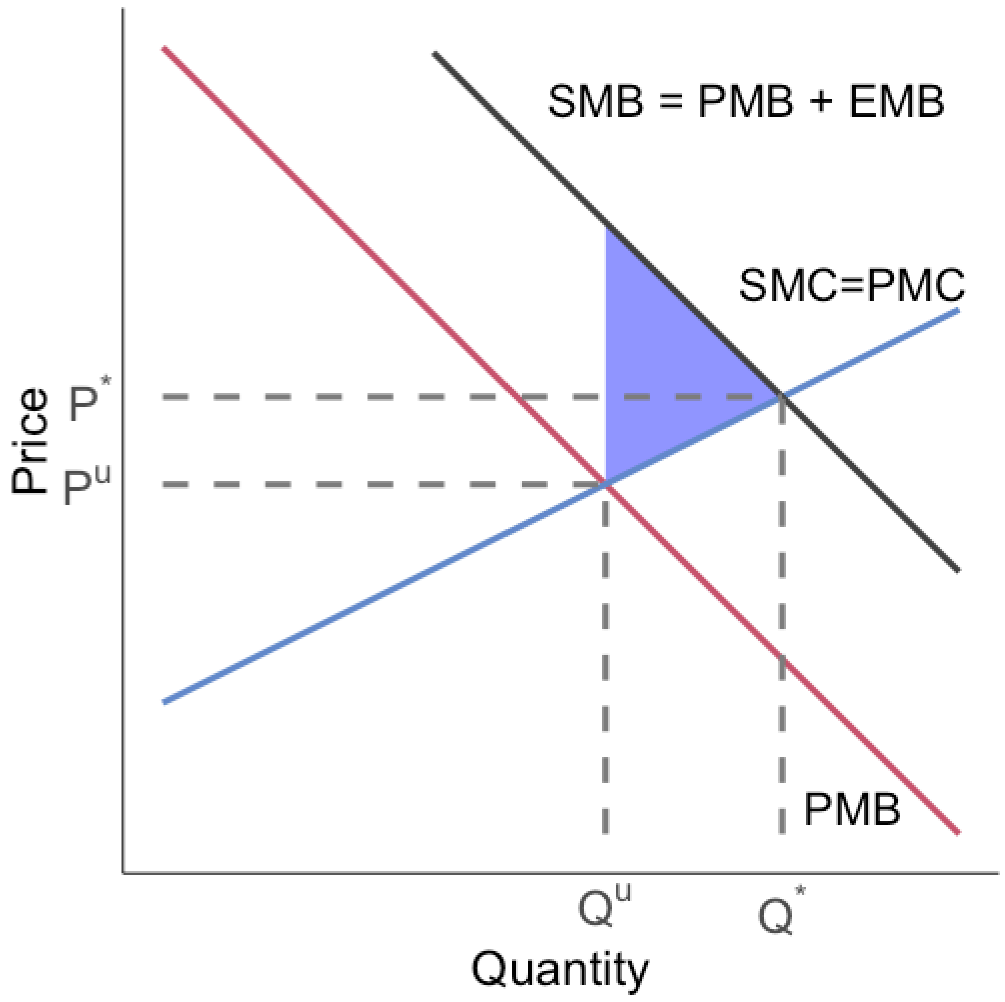
Total SMB - SMC from Q^u to Q^*

Positive externalities: graphical



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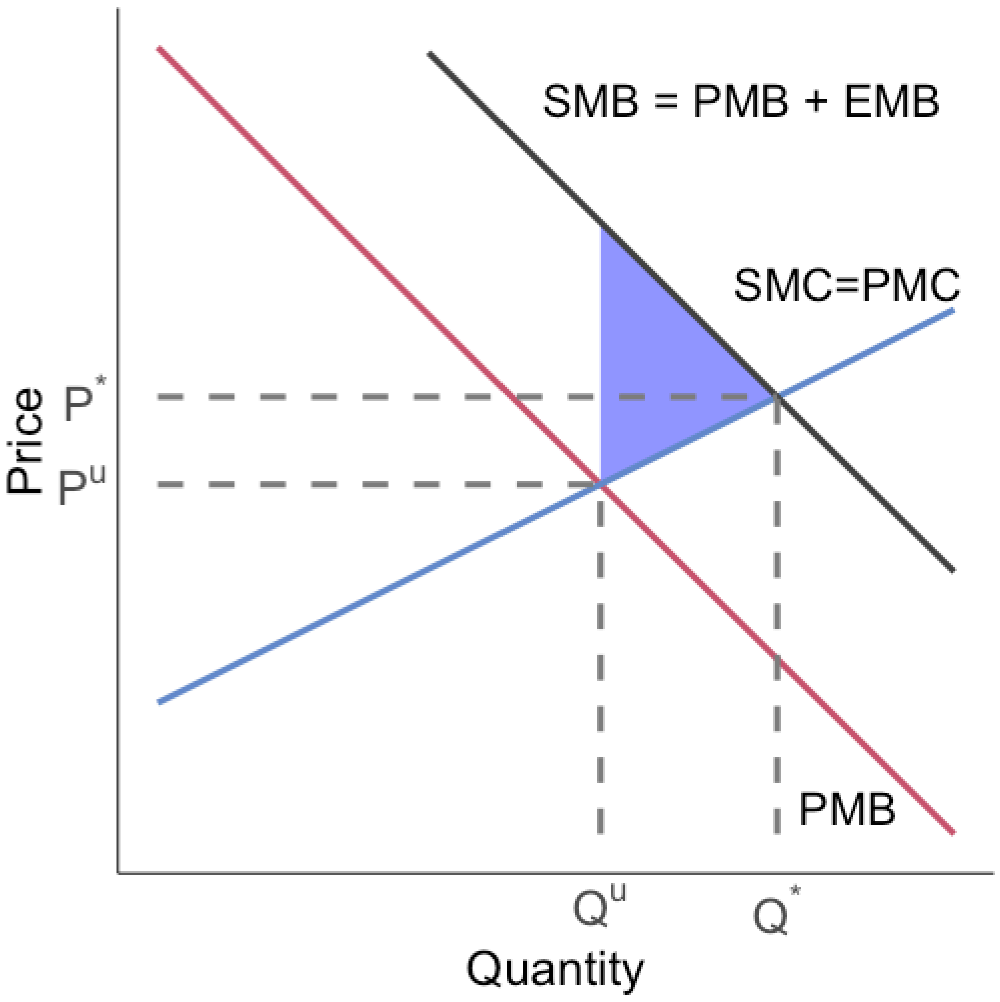
This is the loss to society caused by the externality in the unregulated private market

Positive externalities: graphical

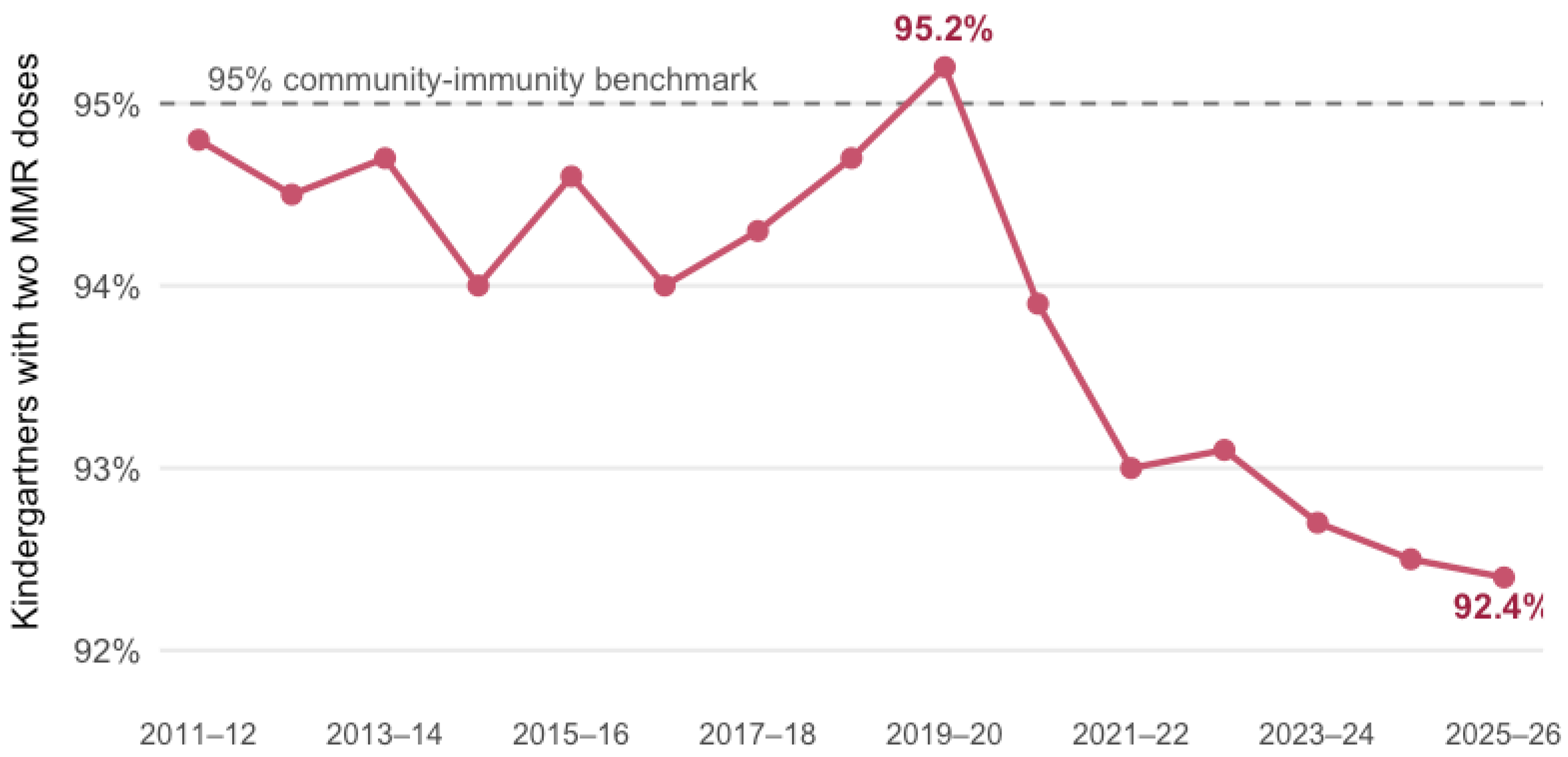


The private market produces too few vaccines

The private actors are not accounting for the external benefits they confer on people who are not in the vaccine transaction (e.g. third parties whose health is affected)

News report: the first U.S. measles deaths of 2026

MMR coverage was near 95% before COVID



Measles returned as vaccination rates fell

On August 25, 2026, Pennsylvania confirmed:

- **Two measles-associated deaths**, both among unvaccinated people
- The state's first measles-associated deaths in 35 years
- 393 confirmed cases across 28 counties during 2026

Two doses of the MMR vaccine provide approximately **97% protection**

Vaccination and infection create externalities

Vaccination can create a positive externality:

- It lowers an individual's probability of infection
- When it also lowers transmission, it protects other people

Infection can create a negative externality:

- Exposure shifts uncompensated risk to other people
- Private choices omit some expected health costs

Efficiency: private choices use PMB and PMC; social choices use SMB and SMC

Why do externalities arise?

Typically one of two reasons:

1. Poorly defined property rights
    - Who owns the right to the air?
2. High transactions costs
    - Hard to bargain over desired air quality with millions of people

Let's conceptualize a model of efficient bargaining using an Edgeworth Box

Why do externalities arise? Edgeworth Box

- Two individuals: A and B
- Two private goods: X and Y

Each individual begins with an initial endowment of each good:

- $A : w_X^A, w_Y^A$
- $B : w_X^B, w_Y^B$

This gives us a total endowment:

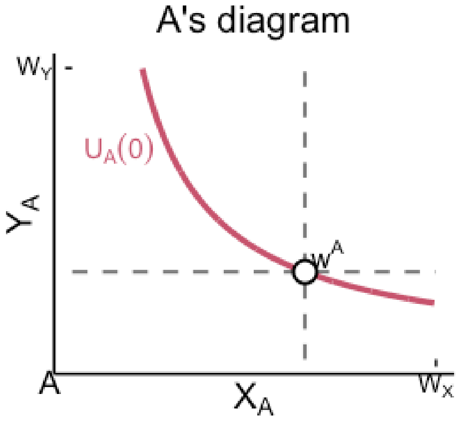
- $W_X = w_X^A + w_X^B$
- $W_Y = w_Y^A + w_Y^B$

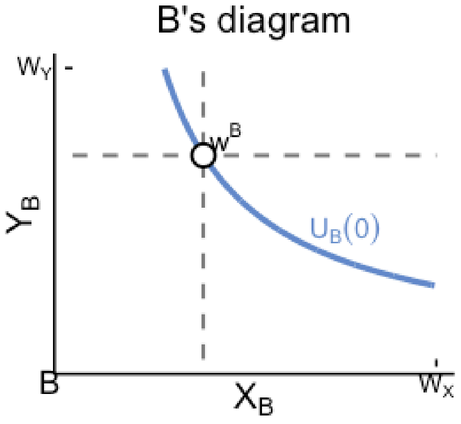
Why do externalities arise? Edgeworth Box

Each person starts with an ordinary consumption diagram

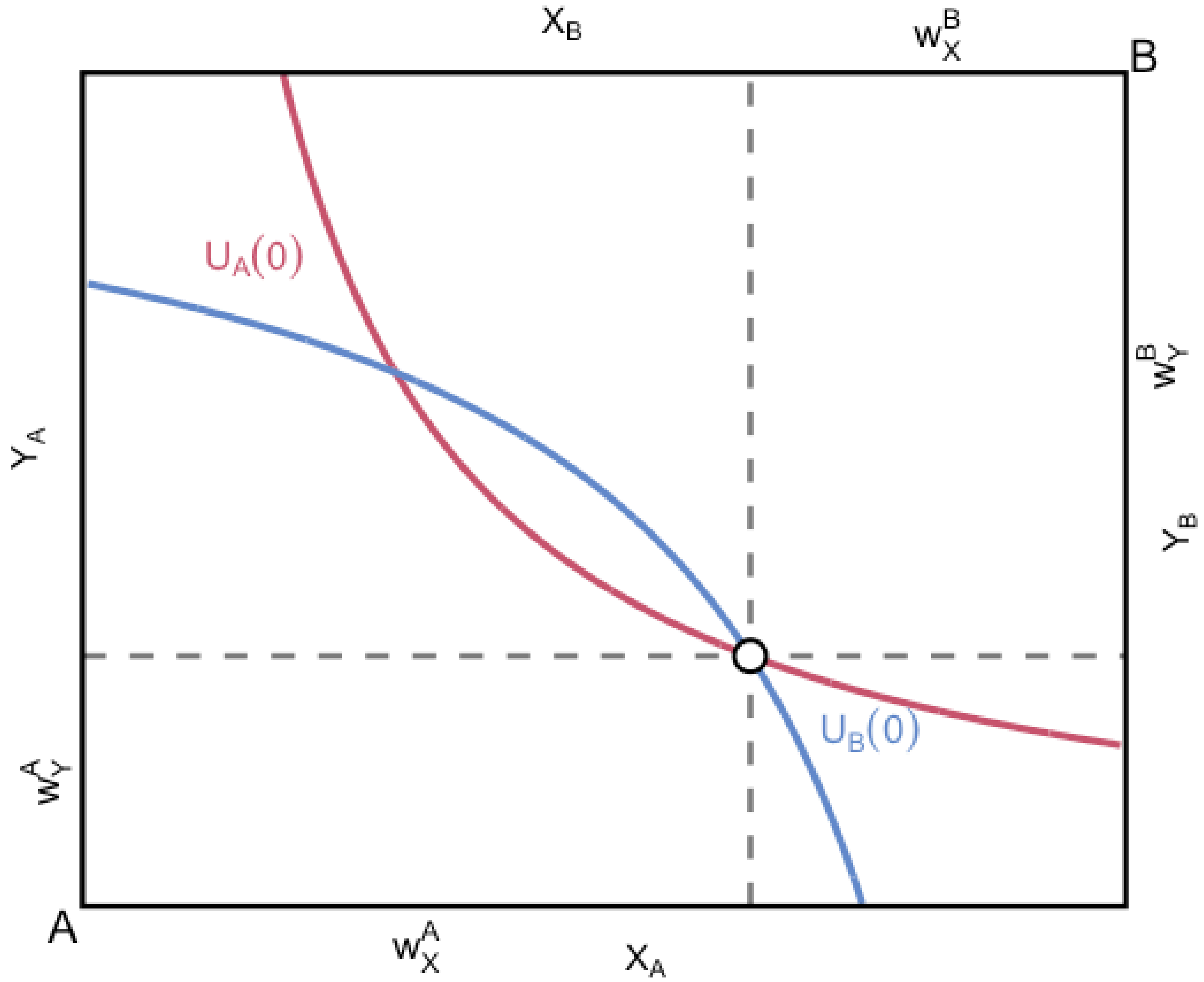
Rotate B's diagram 180 degrees and place it over A's so the endowment points coincide

The result is a box with width W_X and height W_Y





Why do externalities arise? Edgeworth Box



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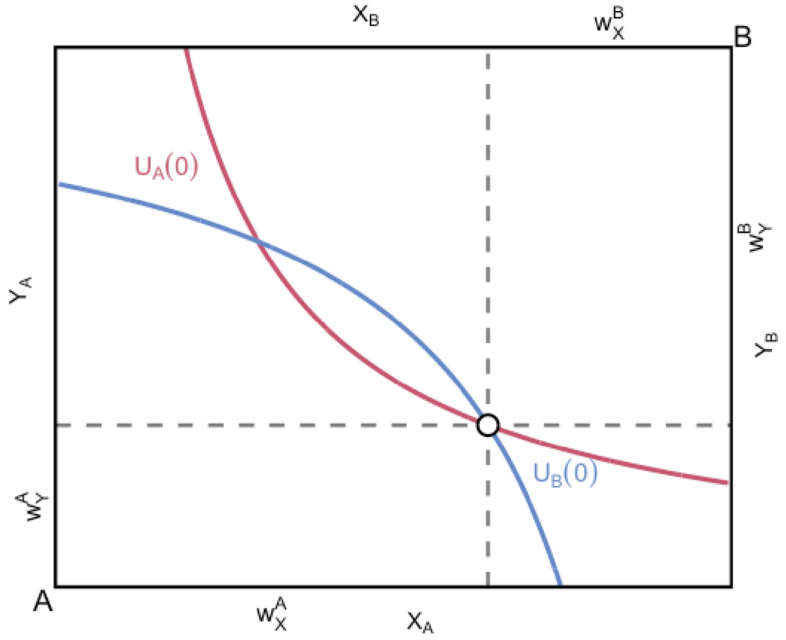


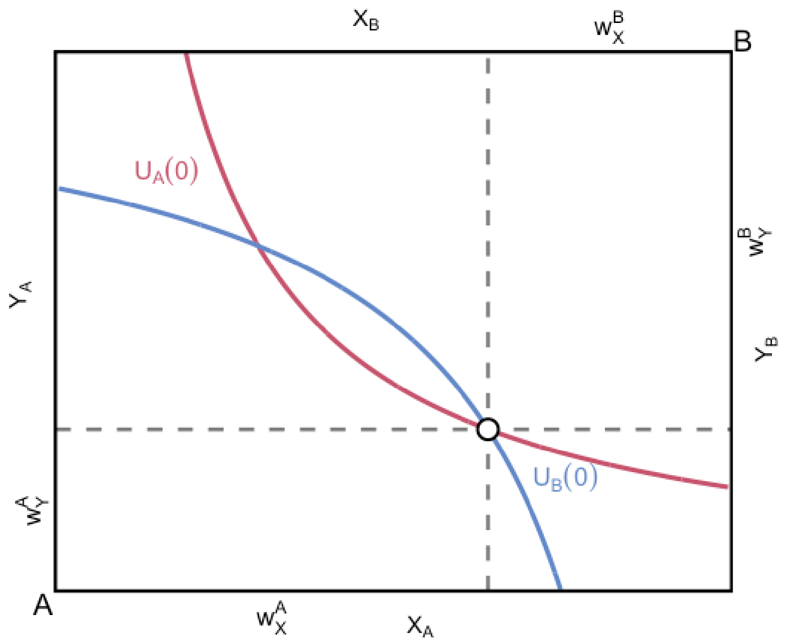
Total vertical distance is W_Y

Total horizontal distance is W_X

Initial endowment is given by the empty circle

Initial indifference curves for A and B are $U_A(0)$ and $U_B(0)$

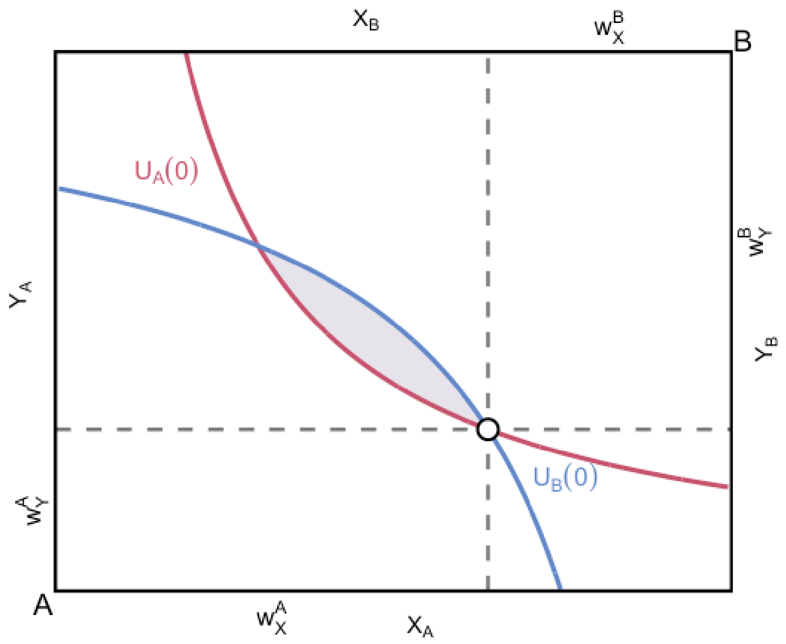
Why do externalities arise? Edgeworth Box



Is there a possible Pareto improvement?

e.g. can we make both A and B better off?

Why do externalities arise? Edgeworth Box

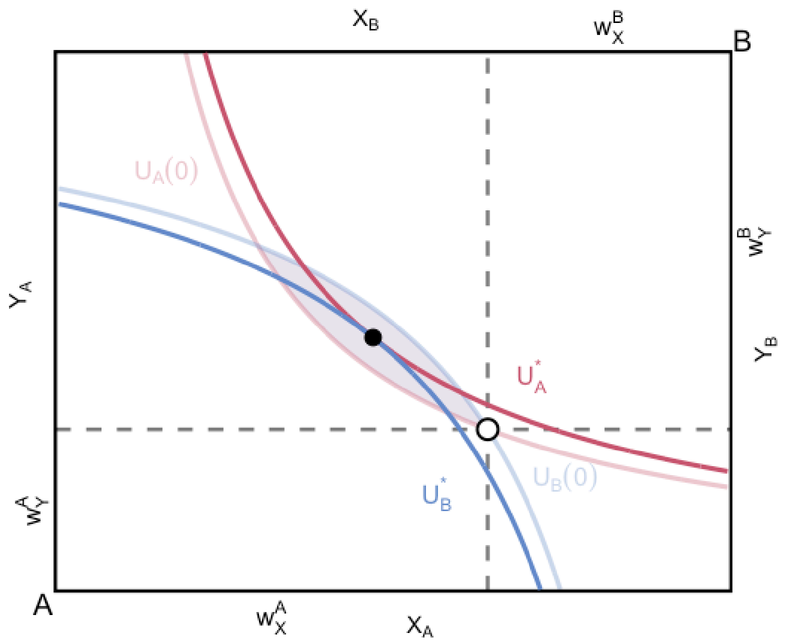


Yes!

The shaded lens is the set of Pareto improvements

If we move anywhere in the lens of their initial indifference curves we have a Pareto improvement

Why do externalities arise? Edgeworth Box

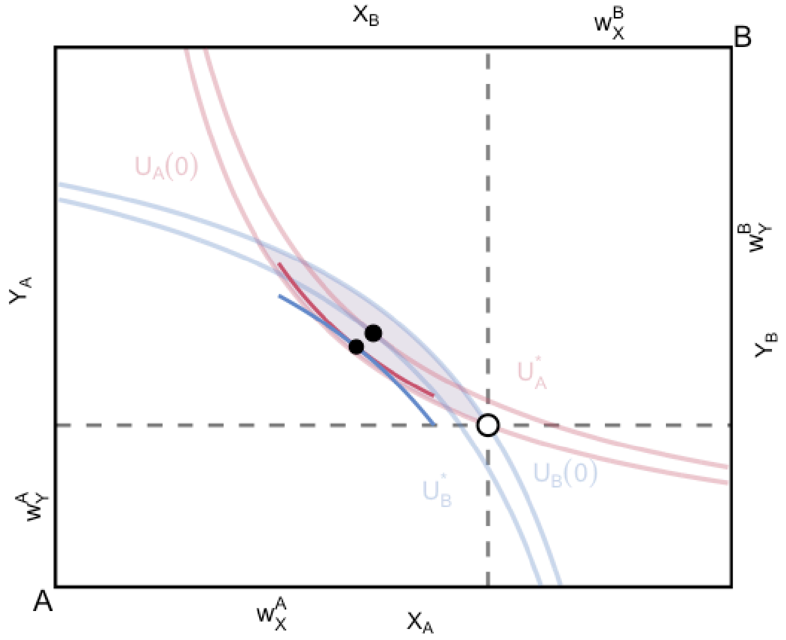


Only a subset of the lens is Pareto efficient

At these allocations no further Pareto improvement is possible

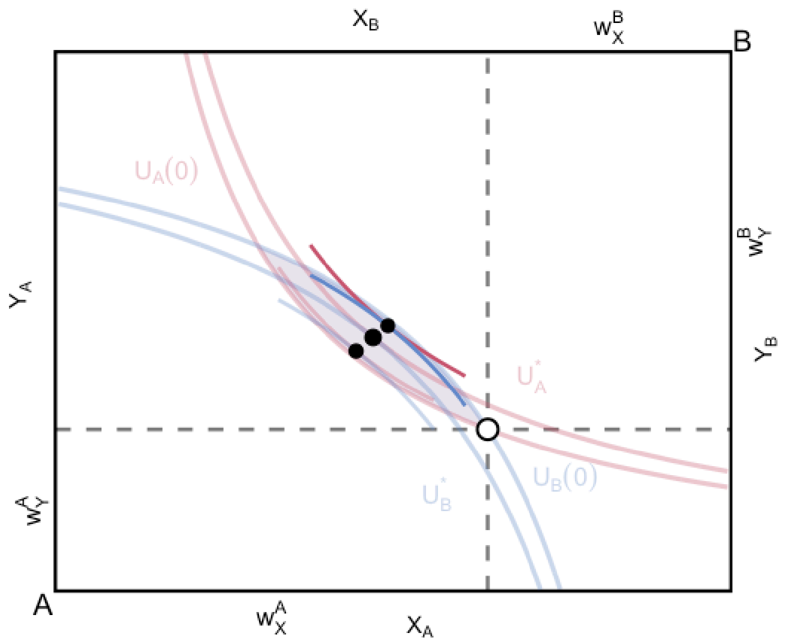
This is where the indifference curves are **tangent** to one another, like the filled-in point

Why do externalities arise? Edgeworth Box



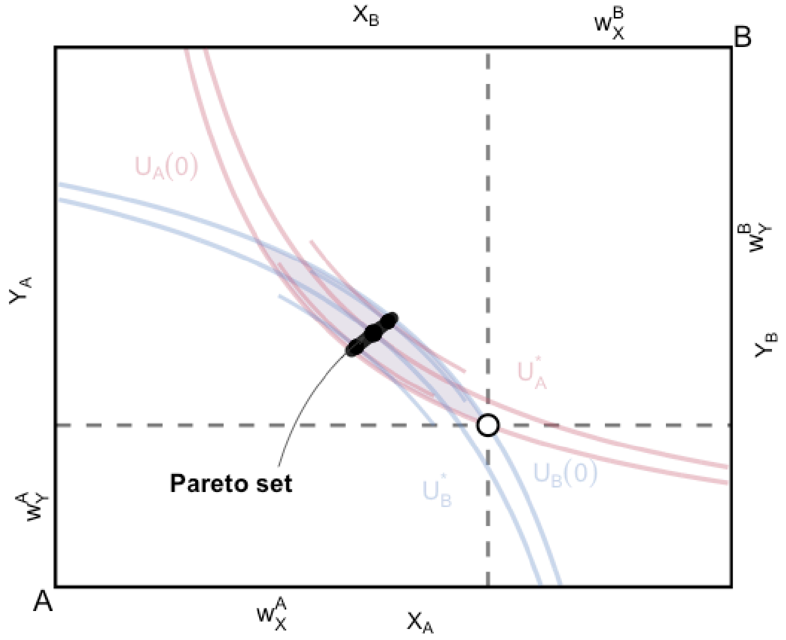
Another tangency gives another Pareto-efficient allocation

Why do externalities arise? Edgeworth Box



More tangencies trace out the Pareto-efficient subset of the lens

Why do externalities arise? Edgeworth Box



This segment is the set of Pareto-efficient allocations reachable by voluntary trade from the endowment

Why do externalities arise? Edgeworth Box

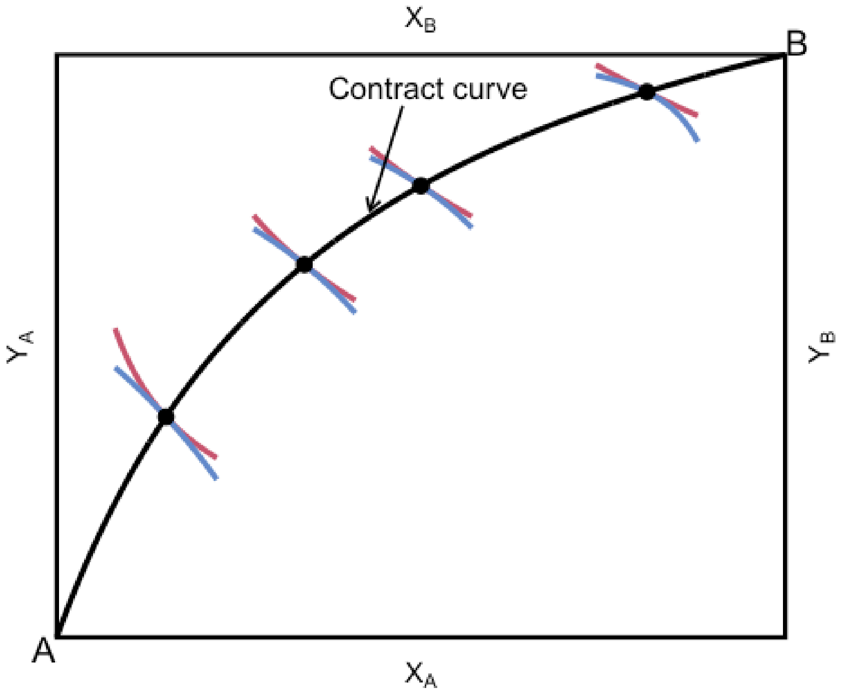
In a properly functioning market:

- The endowment point is well-established
- A and B can trade X and Y to some Pareto improving point
- They continue trading until they achieve a Pareto optimal allocation
- This allocation lies on the **contract curve**: the line consisting of all Pareto efficient allocations

Why do externalities arise? Edgeworth Box

The contract curve collects every allocation where A's and B's indifference curves are tangent

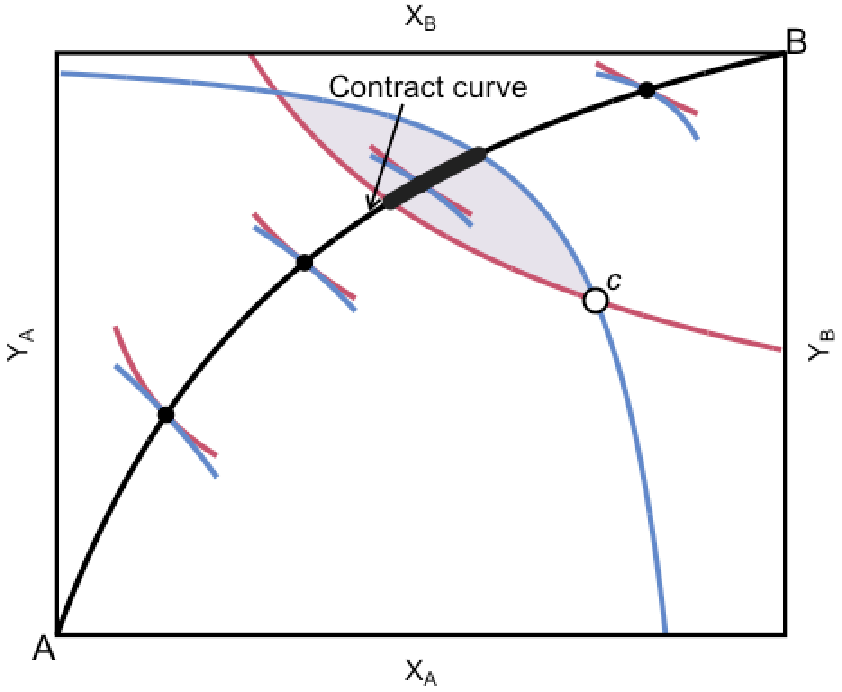
Every point on it is Pareto efficient



Why do externalities arise? Edgeworth Box

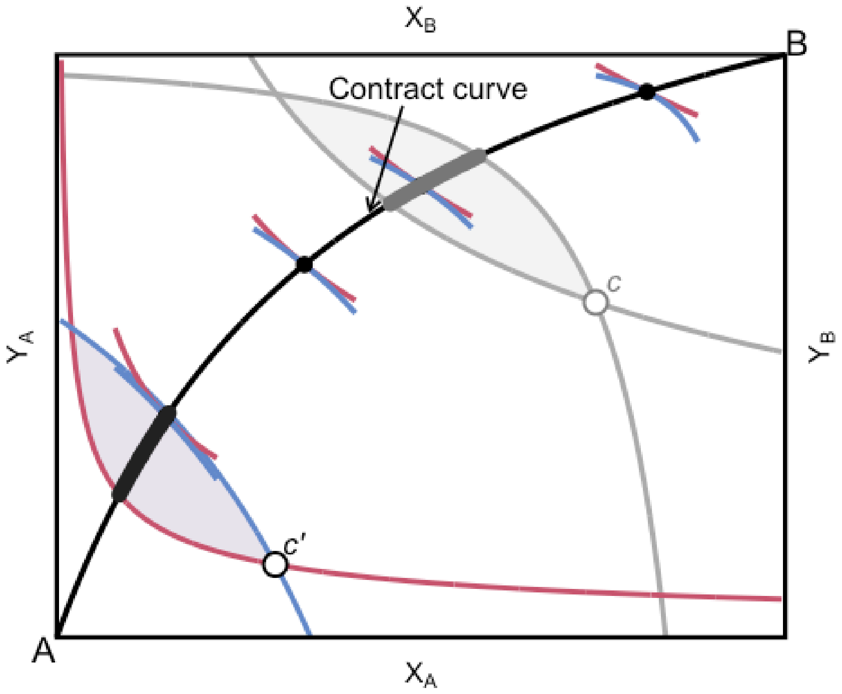
Starting from c, voluntary trade reaches only the thick segment

This is the achievable Pareto set for that endowment



Why do externalities arise? Edgeworth Box

A different initial allocation, c', gives a different achievable Pareto set on the same contract curve



Why do externalities arise? Edgeworth Box

Now suppose Y is not a private good, but a public good/bad, e.g. smoke

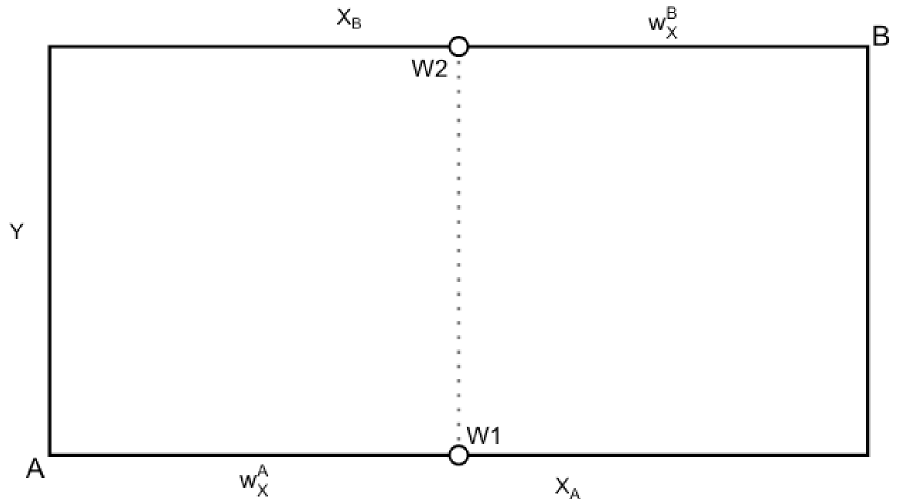
This means that A and B consume the **exact same level of Y**

Unlike our regular Edgeworth Box, now Y increases for **both** A and B as we move to the top of the slide (before Y increased for B as we moved to the bottom)

Suppose that A likes Y, but B does not

Suppose both start off with the same quantity of X

Why do externalities arise? Edgeworth Box

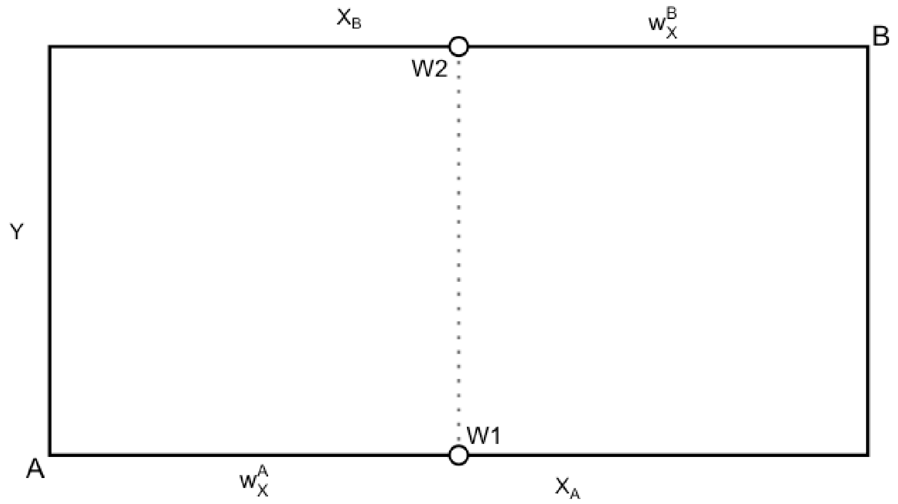


Depending on who has property rights, we either start at:

- W1 (B has property rights)
- W2 (A has property rights)

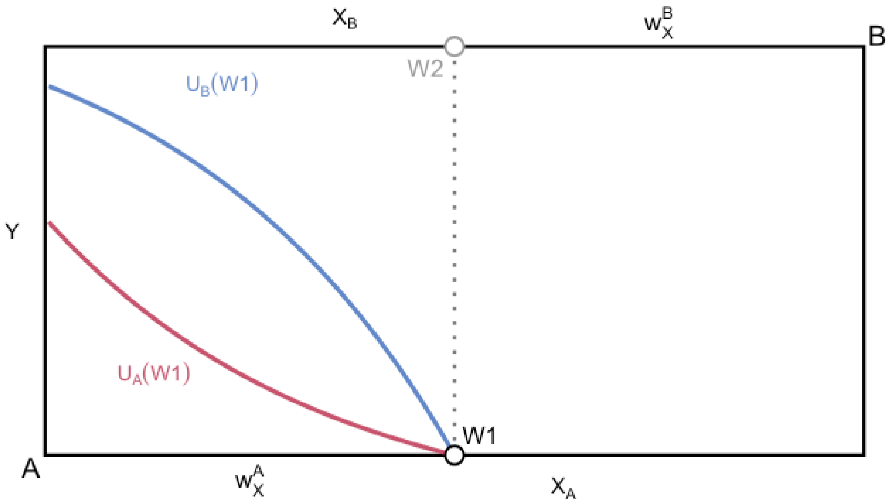
Think about why these are where we must start

Why do externalities arise? Edgeworth Box



Suppose we start at W1, what happens?

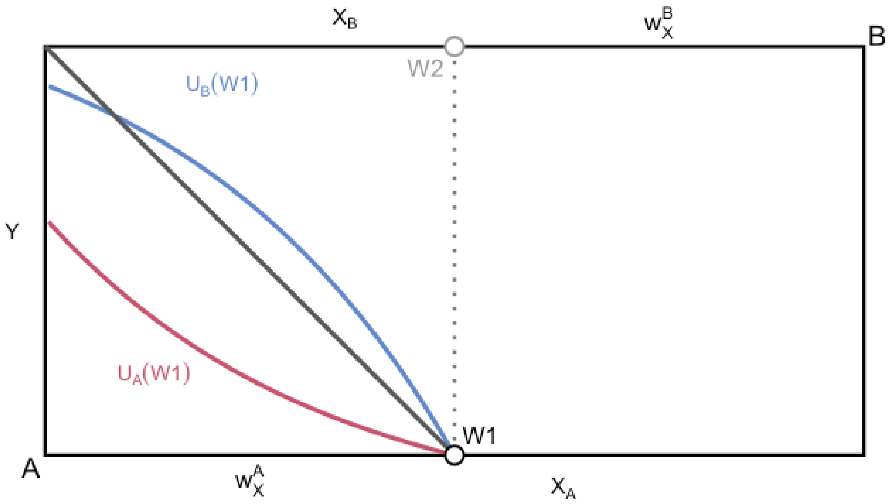
Why do externalities arise? Edgeworth Box



Suppose we start at W1, what happens?

A wants to have more Y, but this imposes a cost on B

Why do externalities arise? Edgeworth Box

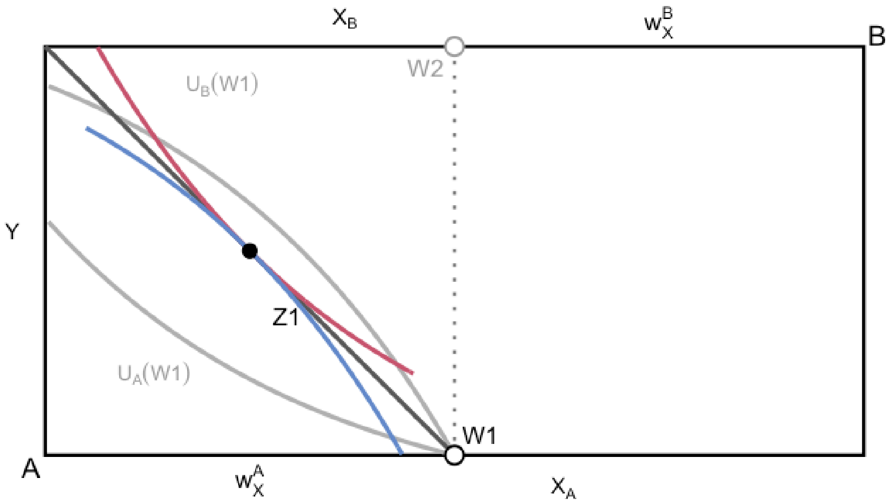


Suppose we start at W1, what happens?

A wants to have more Y, but this imposes a cost on B

Therefore, A has to **pay** B to get more Y

Why do externalities arise? Edgeworth Box



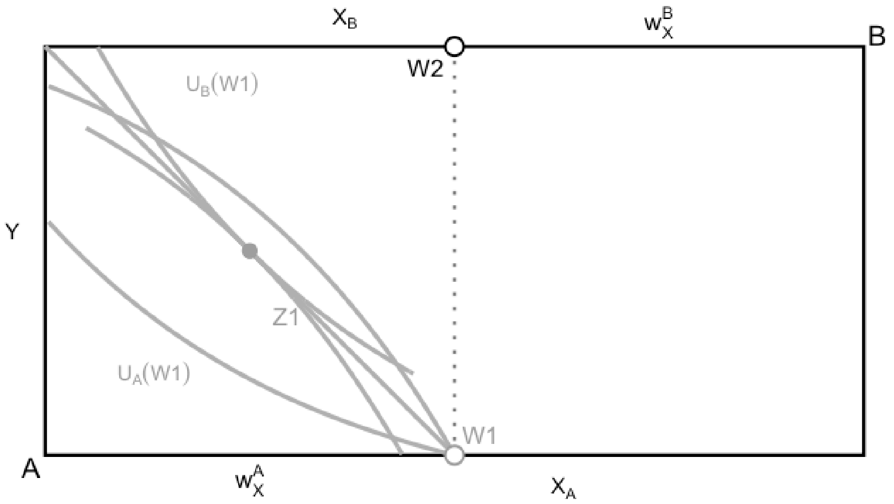
Suppose we start at W1, what happens?

A wants to have more Y, but this imposes a cost on B

Therefore, A has to **pay** B to get more Y

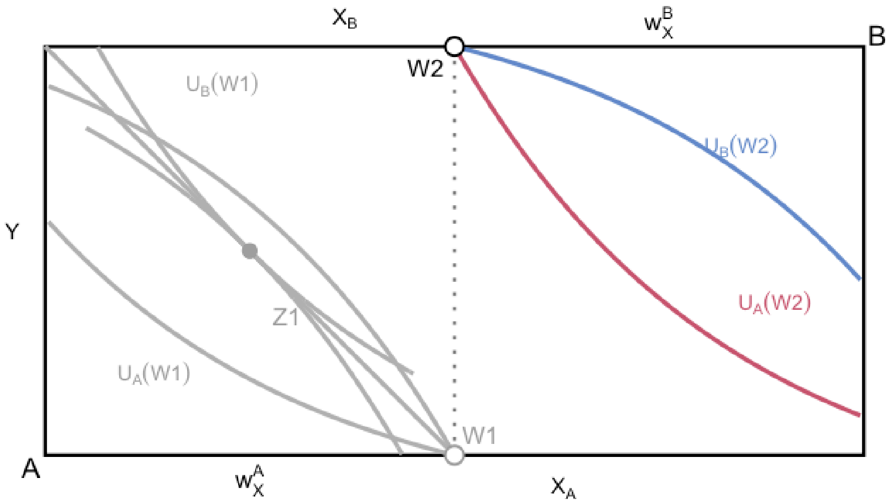
A pays B in units of X, move to Z1, Pareto optimum

Why do externalities arise? Edgeworth Box



Suppose we start at W2, what happens?

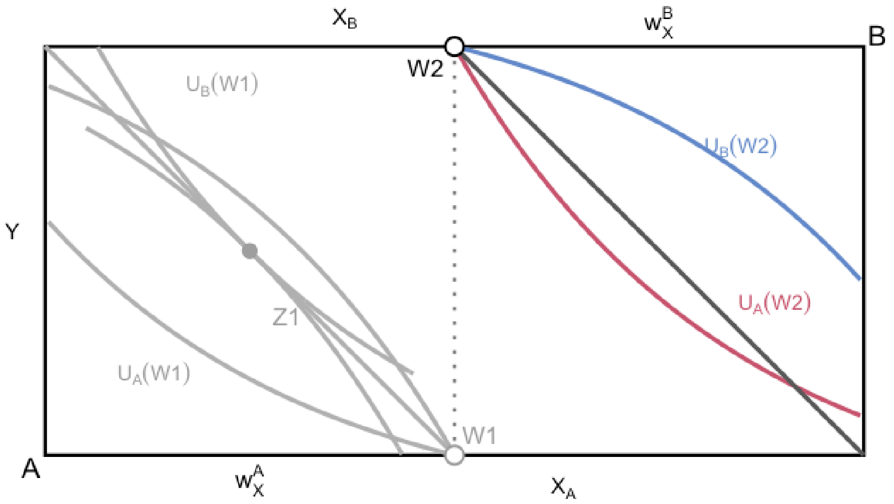
Why do externalities arise? Edgeworth Box



Suppose we start at W2, what happens?

B wants to have less Y, but this imposes a cost on A

Why do externalities arise? Edgeworth Box

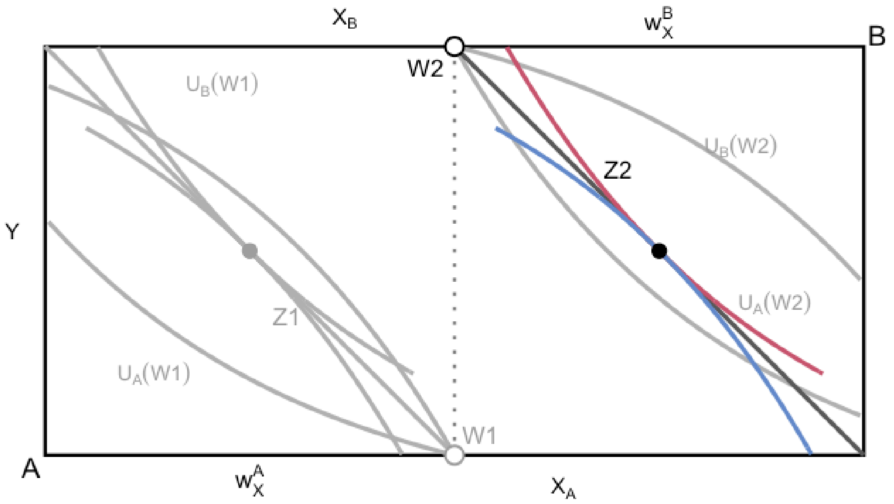


Suppose we start at W2, what happens?

B wants to have less Y, but this imposes a cost on A

Therefore, B has to **pay** A to get less Y

Why do externalities arise? Edgeworth Box



Suppose we start at W2, what happens?

B wants to have less Y, but this imposes a cost on A

Therefore, B has to **pay** A to get less Y

B pays A in units of X, move to Z2, Pareto optimum

Why do externalities arise? Edgeworth Box

In the previous example we were able to achieve the Pareto optimum even with a public good / externality

Why?

1. Property rights were assigned to either A or B

2. Transactions costs were low (didn't have to pay a fee to trade X)

Property rights and externalities

A solution to many externalities is to just assign property rights and let the market do its thing

We'll talk about a few ways that we can assign property rights

Transactions costs and externalities

Now suppose there were many non-smokers

Even if they were assigned the property rights, it might be hard for them to bargain

- Takes a lot of time to find something that works for everyone
- Negotiating over how much X each person gets

The costs of bargaining may exceed the benefits and we end up stuck at W2

Transactions costs and externalities

Road noise: drivers implicitly have property rights to make noise around roads

Even if you prefer quiet, you cannot negotiate a payment with every loud car that might pass

The free-rider problem

Externalities and public goods/bads often exhibit many of the same features

Both are subject to the **Free-Rider Problem**

> A type of market failure that occurs when those who benefit from resources, public goods (such as public roads or hospitals), or services of a communal nature do not pay for them or under-pay

e.g.

- people don't pay their taxes for publicly-provided services
- non-smokers will wait for others to pay in order to reduce smoke

The provision of public goods

Public goods

How do we efficiently provide public goods?

We know:

- Private goods: PMB = PMC $\leftrightarrow$ SMB = SMC
- Goods with negative externalities: PMB = SMC $\leftrightarrow$ SMB = SMC
- goods with positive externalities: SMB = PMC $\leftrightarrow$ SMB = SMC

Suppose we have a public good, e.g. depth of a river for public use

How do we decide the socially efficient depth?

Public goods

Optimal provision is **always** given by: SMB = SMC

What are the SMB and SMC for a public good?

Public goods have two defining characteristics:

Non-rival: one person's use does not reduce the amount available to others

Non-excludable: it is difficult or costly to prevent nonpayers from using the good

Different roles: non-rivalry shapes aggregation; non-excludability causes free riding

Optimal provision of public goods

What does this mean?

When we count up the SMB, we need to add up **everyone's** PMB:

Optimality: $SMB = \sum_i PMB_i = PMC$

If we ignore the fact that public goods are non-rival, we get underprovision of the good

e.g. the free market underprovides clean air, national defense, etc

Modeling the provision of public goods

How do we model public goods?

Here's how to think about it:

For private goods:

Private goods are rival, only one person can consume each unit

At each price, what is the total quantity that is demanded?

At each price, we need to add up quantities

Private goods: we add demand curves horizontally

Modeling the provision of public goods

For public goods:

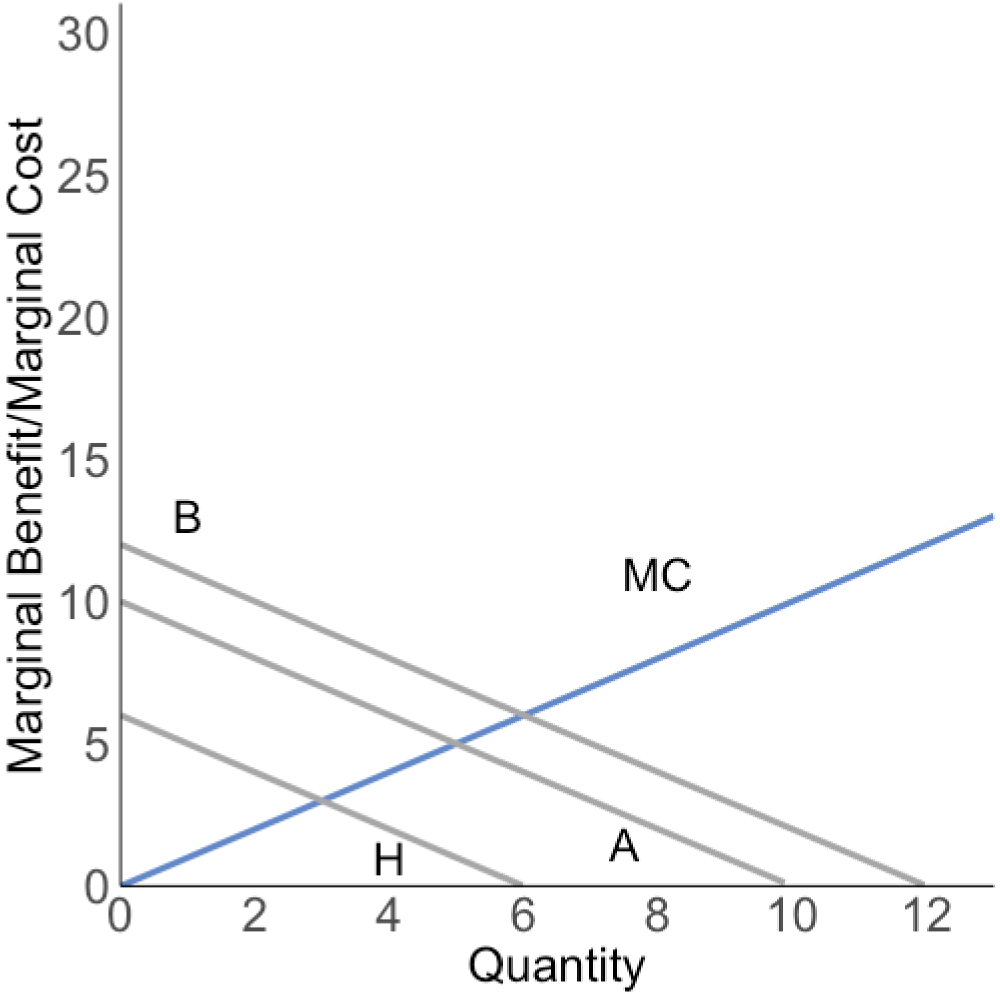
Public goods are non-rival, multiple people can consume each unit

At each quantity, what is the total marginal benefit?

At each quantity, we need to add up PMBs/prices

Public goods: we add demand curves vertically

Public goods: graphical



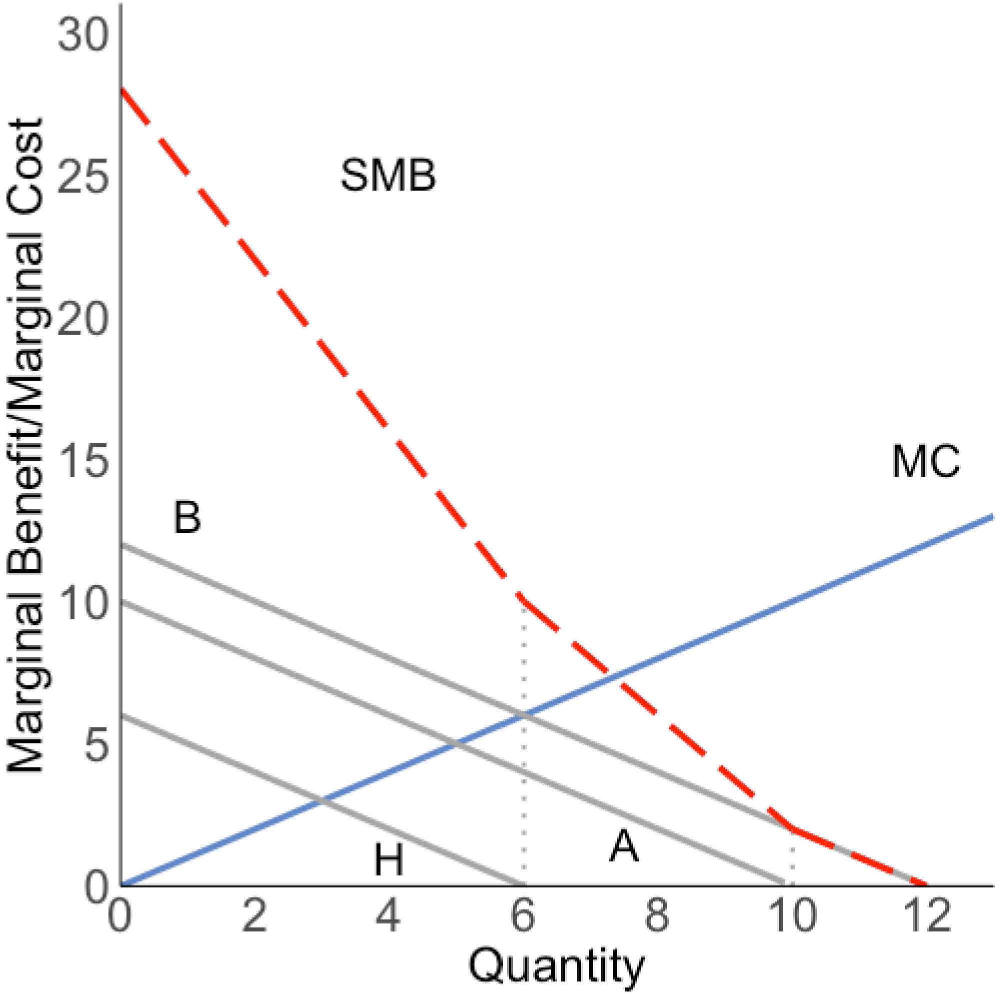
3 different groups: boaters (B), anglers (A), and hikers (H)

Each has a different marginal benefit for water depth:

- Boaters: MB = 12-Q
- Anglers: MB = 10-Q
- Hikers: MB = 6-Q
- MC of provision: MC = Q

We assume MB can't go negative

Public goods: graphical

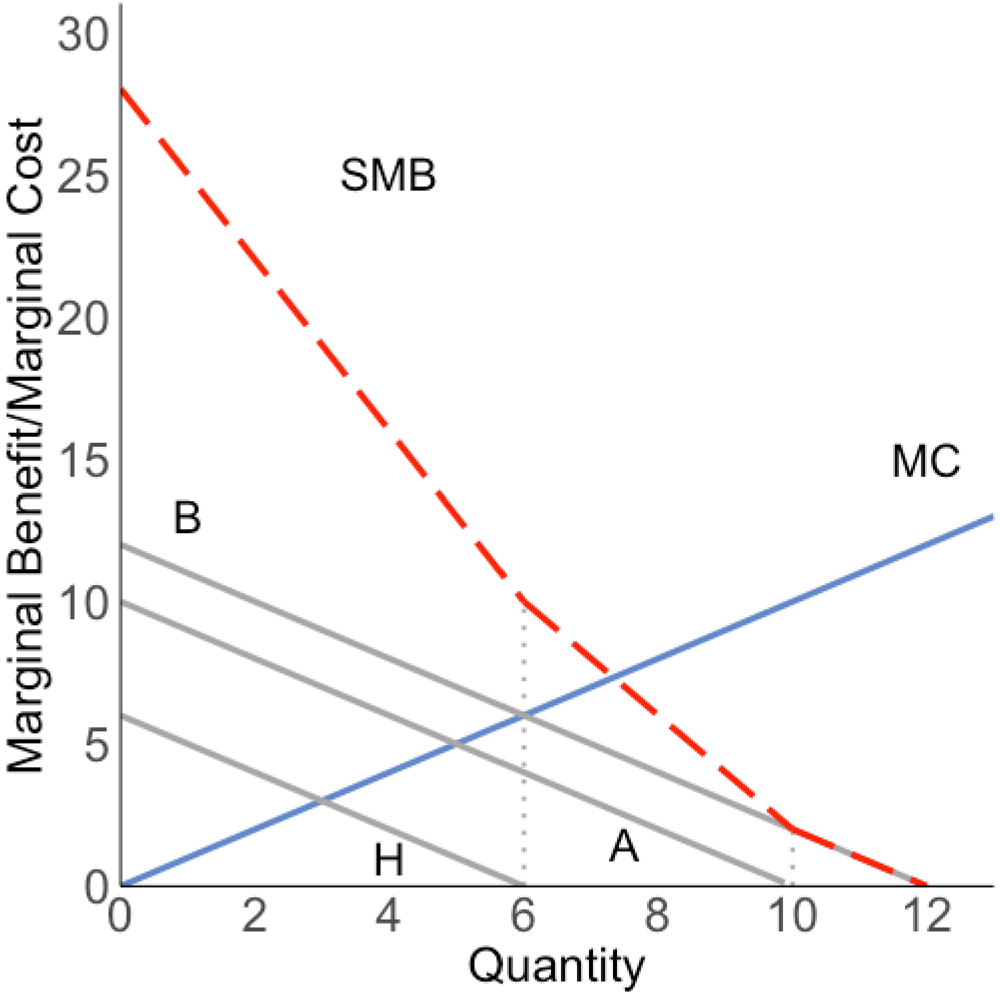


Now we need to aggregate them to get the **social marginal benefit**

We do so by adding up the demand curves vertically:

At each Q, sum the MBs

Public goods: graphical

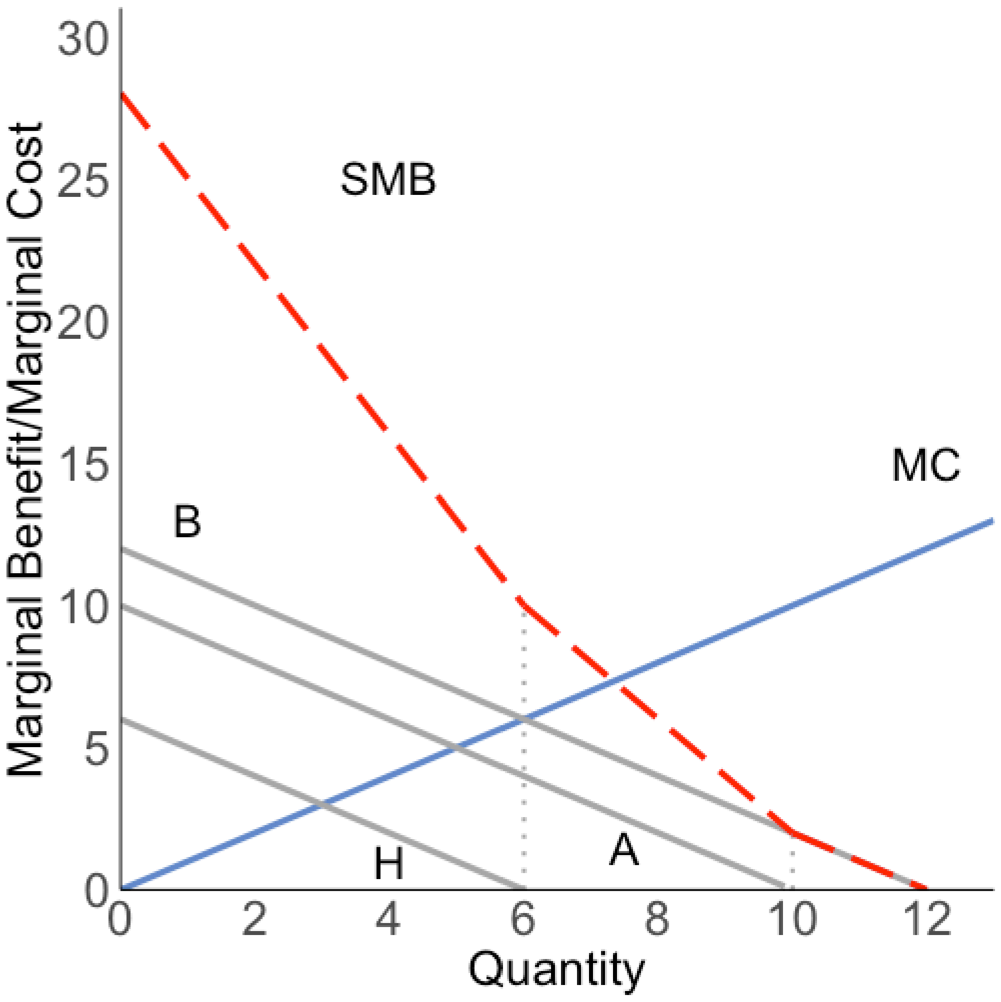


Why is the aggregate demand curve kinked?

Because at each quantity/depth, only certain groups are willing to pay

Kinks are at the dotted lines, where PMBs hit zero

Public goods: graphical

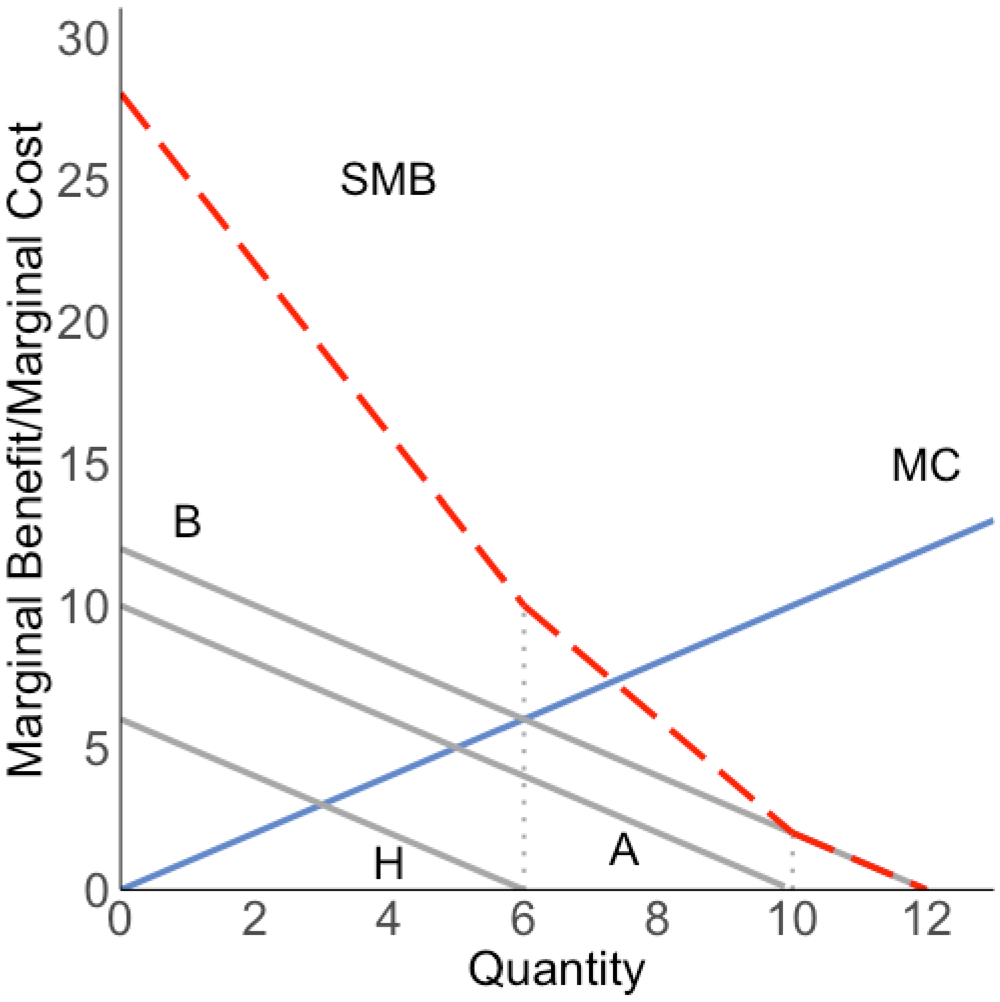


At quantities > 10, only boaters are willing to pay

At quantities > 6 and <= 10, only boaters and anglers are willing to pay

At quantities $Q \leq 6$, all groups are willing to pay

Public goods: graphical



The SMB curve is:

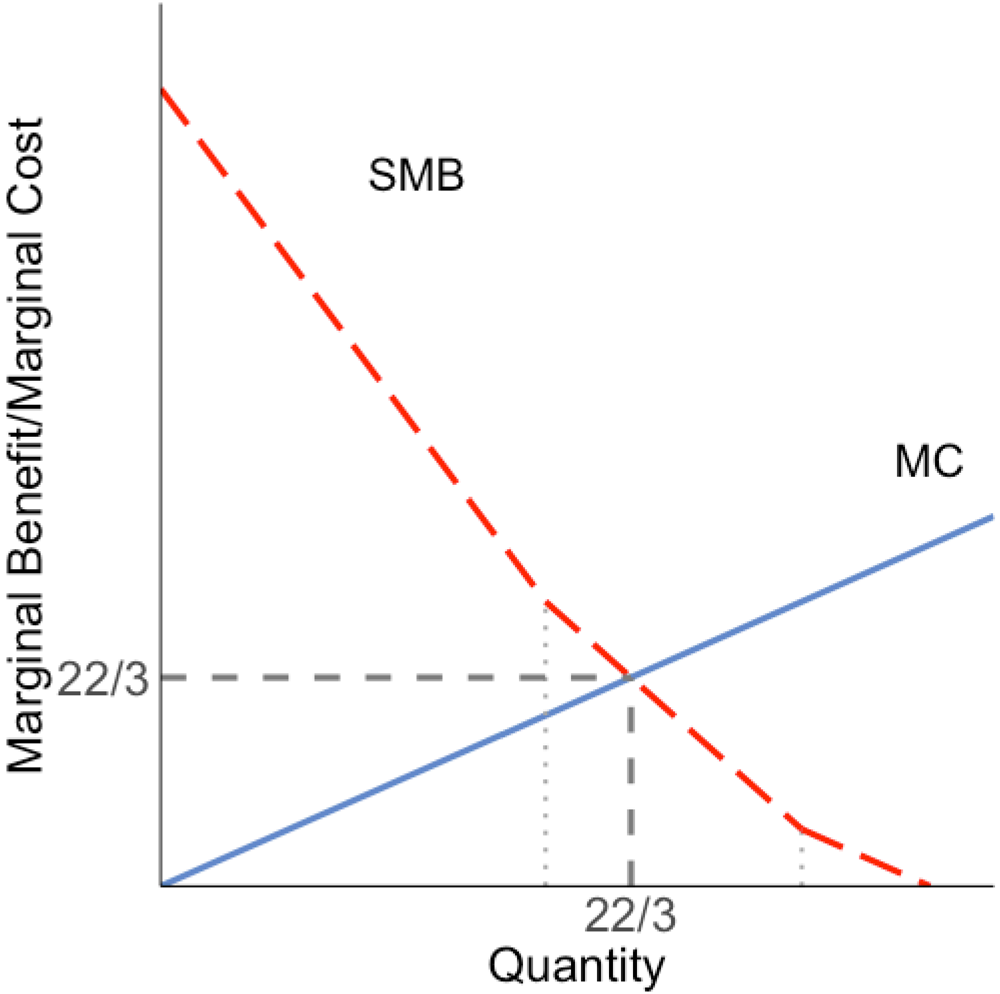
28 - 3Q for Q <= 6

22 - 2Q for 6 < Q <= 10

12 - Q for 10 < Q <= 12

Summing the PMBs over the relevant range of Q

Public goods: graphical



The optimal provision of the public good is where the MC curve crosses the SMB curve

This is across the middle segment

$Q = 22 - 2Q \Rightarrow Q = 22/3$

The optimal quantity of $Q = 22/3$ is greater than the quantity any individual group would be willing to purchase

Public goods financing

The socially optimal quantity is greater than the individual privately optimal quantities

This means that the MC of provision, MC = 22/3

Is this the price the groups pay?

No! It is greater than any individual group is willing to pay

If the government is able to provide the good, how does it finance the cost raising the river depth above zero?

Public goods financing

It can charge each group a constant share of this price

What share is everyone charged?

Lindahl pricing: charge each group equal to their marginal benefit

Boaters pay: 14/3

Anglers pay: 8/3

Hikers: free

Notice that the prices sum to the marginal cost!

Since the good is non-rival, this is enough to finance the cost

Public goods financing

What is a key problem with Lindahl pricing?

People can lie about which group they're in

Anglers might say they're hikers

It requires perfect information on behalf of the regulator